

The Trade-Off Between Decarbonization and Reshoring

Evidence from the French EV Environmental Score Reform ^{*}

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Abstract

Governments increasingly tie clean-technology subsidies to industrial aims. In 2024, France conditioned its electric-vehicle (EV) purchase bonus on a life-cycle environmental score, instantly disqualifying models accounting for about one-third of EV sales. Linking exhaustive registration records to assembly plants and battery origins, we exploit this sharp eligibility shock in a difference-in-differences design. We then combine these estimates with original survey data on second choice to estimate a nested-logit demand system. The reform redirected demand toward European supply chains but reduced EV penetration by 0.4 percentage points. The resulting increase in tank-to-wheel emissions roughly offsets manufacturing-stage CO₂ gains. A budget-neutral variant recycling fiscal savings into larger bonuses for eligible models restores electrification and turns the net CO₂ effect positive, at a cost of about €4,200 per reshored battery in consumer surplus. Accounting for producer surplus accruing to domestic shareholders and corporate tax revenue, this cost falls to approximately €1,600 per battery. Our welfare analysis further highlights that the reform effectively converts largely untaxed carbon (manufacturing-phase emissions, outside the scope of carbon pricing) into heavily taxed carbon (driving emissions, taxed at approximately €250/tCO₂ through fuel excise). Accounting for this pre-existing tax wedge dramatically improves the fiscal efficiency of the reform.

Keywords: car industry, subsidies, environment

JEL Codes: H23, Q58, L62, D12, L52

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1 Introduction

Passenger cars are a natural focal point for climate policy: they alone account for roughly 10% of global transport CO₂ emissions (Sen and Miller, 2023). In many advanced economies, this proportion is higher and has grown during recent decades.¹ To hit mid-century decarbonization targets, more than twenty-five countries (and many sub-national jurisdictions) have announced that sales of new Internal Combustion Engine (ICE) cars must cease by 2035 or earlier. These phase-out pledges are reinforced by generous purchase incentives that narrow the price gap between battery-electric vehicles (EVs) and their internal combustion engine (ICE) counterparts.

The transition, however, is not merely environmental—it is also industrial. Batteries and final assembly represent a large part of the value added of a vehicle. If, as is the case for ICEs, these stages tend to cluster, the shift to EVs risks being accompanied by a large spatial shift in production. China’s rapid ascent sharpens that concern: By 2023, China already represented 62% of the world’s EVs and nearly 80% of battery cells, far above its share in the traditional ICE market. Policymakers have therefore begun to tie EV subsidies to localisation or domestic-content rules, the most prominent example being the U.S. Inflation Reduction Act (Allcott et al., 2024).

Then, there is the CO₂ emissions trade-off. Building an EV typically embeds 2 to 4 tons of CO₂-equivalent more than building a comparable ICE car, mainly because of battery manufacturing and aluminum-heavy bodies (IEA, 2024). On the road, the ordering reverses: EVs emit virtually no CO₂, whereas tank-to-wheel emissions of ICE cars are roughly 120–180 grams of CO₂ per kilometer.² The “break-even” distance at which an EV’s life-cycle emissions fall below those of an ICE is therefore around 30,000–50,000 km, depending on how clean the electricity mix is (IEA, 2024). Policies that screen EV models on life-cycle (including the manufacturing-phase) emissions curb high-emitting production today but this can be at the price of slowing down the long-run decarbonization of the fleet if they restrict the set of eligible EVs to such an extent that electrification of the fleet stalls.

France’s 2024 reform illustrates this tension well. Only EVs for which manufacturing emissions fall beneath a fixed life-cycle threshold continue to qualify for the purchase bonus, a rule that instantly disqualified models accounting for about 30% of EV sales in 2023. Because life-cycle intensity varies strongly with the location of assembly and battery production and (to a lesser extent) with transport costs, the measure also acts as an filter that favors regional supply chains. This paper asks whether such Life-Cycle-Assessment (LCA)-based incentives can *simultaneously* speed up fleet decarbonization by steering demand toward lower-emission production while favoring domestic producers, and at what economic cost.

We assemble a new monthly panel that links every passenger-car registration in France to its place of final assembly and—when the vehicle is electric—to the country (or region) in which its battery cells were produced. Two sharp policy changes in January 2024 provide quasi-experimental

¹For example, while France’s overall greenhouse-gas emissions fell by 20% between 1990 and 2019, transport’s emissions stagnated and its share of overall emissions went up from 27% to 32% over the same period. Within that sector, passenger cars represent over 50% of France’s transport-sector CO₂ footprint in recent years (CITEPA, 2023), implying that passenger cars represent at least 16% of overall emissions.

²Recent works in France and the EU (European Commission, 2024) show that WLTP norms tend to underestimate the real world emissions, with real world emissions exceeding the WLTP norm by 20% for gasoline and diesel engines and 350% for plug-in hybrids.

price variation: first, and most importantly, the life-cycle “environmental-score” filter mentioned above disqualified a large share of EV models from the French purchase “bonus”. Second, the pre-existing CO₂ “malus” on internal-combustion cars (ICEs) was increased for high-emitting models. By comparing within-segment market shares of treated vs untreated vehicles before and after each reform in a difference-in-differences design, we are able to assess how responsive demand is to policy-induced price changes.

To translate those reduced-form findings into substitution patterns usable for the whole sector, we embed those shocks in a three-level nested-logit model. Households first decide among not buying, purchasing an ICE, or purchasing an EV; conditional on the powertrain, they choose a vehicle segment and then a model. The correlation parameters that govern within-nest substitution are identified jointly from the policy variation and from a French “second-choice” survey (May 2026) that reports whether EV buyers would remain in the EV nest were their preferred model unavailable. Solving the model—either under monopolistic competition with constant markups or under multiproduct Bertrand pricing—allows for simple “exact hat-algebra” counterfactuals.

Our new dataset shows the pre-reform dominance of Chinese supply chains: in 2023, China supplied about 30 % of French EV registrations and nearly 55 % of battery cells. Our DiD estimates confirm strong price sensitivity and dynamic specification shows no evidence of differential pre-trends, thus lending credibility to the identifying assumption. Losing the bonus cuts sales of the affected EVs by roughly 60 %, an own-price elasticity of $\varepsilon \approx -4$; a one-thousand-euro rise in the malus reduces sales of targeted ICE models by about 20 %, implying elasticities $\varepsilon \approx -5$.

Feeding these responses through the nested logit model allows for a quantification of the trade-offs between reshoring, decarbonization and the consumer surplus that lies at the heart of life-cycle-based industrial policy. Our results show that the environmental-score filter succeeds in redirecting demand toward European assembly plants and battery suppliers, resulting in the additional sales of about 10,168 cars with European-sourced batteries, yet at the cost of a 0.4-percentage-point fall in overall EV penetration. Consumer surplus declines by about €240 million but this is more than offset by €228 million in fiscal savings. The net CO₂ effect is (slightly) negative once higher tank-to-wheel emissions are accounted for. A budget-neutral alternative that retains the life-cycle filter but recycles the savings into a larger bonus for the remaining eligible EVs restores electrification and generates net benefits in terms of emissions. It increases the number of European-sourced batteries by around 14,320 units, yet still costs roughly €4,200 in consumer surplus for every additional battery sourced in Europe. When producer surplus accruing to French citizens is taken into account, this cost falls to approximately €1,600.

Our welfare analysis sheds light on an asymmetry in how the two types of emissions affected by the reform are priced. The reform reduces manufacturing-phase CO₂—a source that is largely untaxed, as no carbon price applies to battery production or vehicle assembly outside the EU Emissions Trading System—while increasing tank-to-wheel emissions from driving, which are taxed through France’s fuel excise (TICPE) at an implicit carbon price of approximately €250 per ton,³

³The TICPE rate on petrol (SP95) is €0.68 per litre, corresponding to approximately €300/tCO₂ (2.28 kg CO₂/litre). For diesel (B7), the rate of €0.59 per litre implies approximately €225/tCO₂ (2.64 kg CO₂/litre). Weighting by the approximate diesel-petrol split among new cars in France (55%–45%) yields an average of roughly €260/tCO₂. The

well above the social cost of carbon of €200 used in our calculations. This wedge between taxed and untaxed emissions is particularly consequential in France, where the electricity grid is largely decarbonized (nuclear), making tank-to-wheel emissions from EVs exceptionally low. In such a context, as Holland et al. (2016) emphasize, the environmental benefit of driving electric is large—but the manufacturing-phase emissions embedded in vehicle and battery production become the dominant margin of differentiation across models. By conditioning subsidies on life-cycle emissions, the reform partially corrects the misallocation arising from the unpriced manufacturing margin—though it does so at the cost of modestly expanding the already-overtaxed driving margin. The reform effectively converts untaxed carbon (manufacturing) into overtaxed carbon (driving)—a second-best correction that would be unnecessary if manufacturing emissions were priced directly, for instance through a carbon border adjustment mechanism for batteries. This mechanism is first-order for the welfare assessment: the Average Value of Public Funds falls from 1.07 to 0.75 once fuel-tax revenue is included, meaning that the apparent fiscal efficiency of the reform is largely driven by pre-existing wedges in carbon pricing across emission sources.

Relationship with the literature. The first strand of work that our paper speaks to evaluates how demand-side incentives shape consumers’ adoption of lower-carbon vehicles. Early quasi-experimental studies exploited a variety of policy changes and found sizable but heterogeneous elasticities of demand for hybrids and EVs—for example, Chandra et al. (2010), Klier and Linn (2015), and Yan and Eskeland (2018). More recent evidence from large-scale programs shows that elasticities remain high when incentives are targeted at lower-income households or bundled with non-price perks: see Muehlegger and Rapson (2022) for California. In comparison with Muehlegger and Rapson (2022), we tend to find larger reduced-form own price elasticities for EVs, which might reflect the fact that our policy-induced variation in price affected vehicles that present very similar attributes as far as the final consumer is concerned. Our paper is most closely related to Allcott et al. (2024) which focuses on the short-run impact of an EV incentive program (the US Inflation Reduction Act, IRA, implemented by president Biden at the end of 2022) which impacted differentially domestic and foreign car makers. We build on their approach by estimating a similar nested model, but use French survey moments rather than the US survey moments in Allcott et al. (2024). We depart from them in several ways. Where Allcott et al. (2024) analyze an explicit domestic-content rule under the IRA, we examine a reform that conditions subsidies on a vehicle’s cradle-to-gate CO₂ footprint and show its *de facto* domestic / European bias. Because the French reform hit a slack market without the inventory rationing that characterized the car market until mid 2023 (reference), our difference-in-differences design isolates the pure demand response to losing the bonus rather than a mix of supply and demand shocks. Moreover, we trace the provenance of both final assembly and battery cells for every registration, letting us measure how the policy redirects sourcing along the EV value chain—a stage at the heart of many industrial policy concerns (Head et al., 2024; Barwick et al., 2025). We also depart from them by adopting a somewhat different

OECD’s *Effective Carbon Rates* report (OECD, 2023) documents that road transport faces effective carbon rates an order of magnitude higher than those in industry and manufacturing across OECD countries. On the interaction between environmental and pre-existing taxes in a second-best setting, see Bovenberg and Goulder (1996).

approach to identification and estimation. In the spirit of Berry et al. (2004), we show that second choice data can be used sequentially to identify the nested parameters, without making assumptions about the structural error term nor estimating the price parameters. Furthermore, applying insights from “exact hat algebra” to the nested logit, we build moments as a function of the nest parameters and observed shares only, which eases and speeds up estimation substantially.⁴ Our contribution to this literature is to examine a uniquely sharp, LCA filter that removed a purchase incentive from about one-third of the French EV market overnight, thus providing clean quasi-experimental variation. We also contribute by documenting clearly the implication of the policy on the sourcing of the cars but also of the batteries—a component at the center of key industrial policy concerns.

A second, fast-growing literature studies industrial-policy dimensions of the battery electric vehicle transition. Recent models highlight how learning curves, network effects, and strategic trade considerations can justify production-side support for “green” industries (Aghion et al., 2024). Empirical work has focused on place-based subsidies for battery plants (Criscuolo et al., 2023), the reaction of location choices to demand subsidies contingent on local production (Head et al., 2024), and the export consequences of China’s EV push (Barwick et al., 2025). Yet few papers quantify the trade-offs between domestic value-added and carbon abatement when purchase subsidies themselves become conditional on provenance or on LCA scores. By linking exhaustive French registration data to assembly plants and battery origins, we are able to trace how such provenance-based rules redistribute demand across firm nationalities and stages of the value chain. Our model focuses on the short to medium run impact of the policy taking location and sourcing decision by firms as given as well as overall productivity.

Our empirical strategy combines a causal research design with a structural demand model. We first use a difference-in-differences design to estimate the causal effect of the policy on demand. We then embed these responses in a nested logit demand system to recover substitution patterns and simulate counterfactual market outcomes. Conceptually, this approach separates identification from extrapolation: the DiD design identifies average causal responses, while the structural model imposes economically motivated restrictions that translate these responses into a full demand system. This interpretation follows recent work connecting causal inference and structural modeling in industrial organization (Chen, 2025).

Organization of the paper. The paper is structured as follows. Section 2 presents the environmental score reform at the center of the paper. Section 3 shows key descriptive statistics on the dynamics of market shares among EVs and of the overall EV share around the reform. Reduced-form results are in section 4. Section 5 introduces the conceptual framework, presents the demand and supply side of the model as well as our approach to identification and estimation. Section 6 carries out a cost-benefit analysis of the reform and simulate a simple, counterfactual policy. Finally, section 7 concludes.

⁴This can be useful if one wants to use bootstrap inference to gauge the degree of statistical precision of the counterfactual (d’Haultfoeuille et al., 2014).

2 Policy background and data

2.1 Policy background

Electrification incentives for cars in France combine a demand-side incentive (bonus) for battery-electric models, a malus tax on CO₂-intensive ICEs, and several ancillary tweaks (social leasing, fleet rules). In 2024 a new environmental score (ES) fundamentally redefined bonus eligibility by imposing a life-cycle-emission threshold, while the malus schedule steepened sharply. This section documents the three building blocks—bonus/ES, malus, and ancillary changes. In the final subsection (2.1.4), we explain how we translate them into the tax-subsidy variables used in the empirical analysis and in the counterfactuals.

2.1.1 The environmental score reform: change in the main EV incentive program

The bonus before the reform. Since 2023, France’s purchase subsidy has been reserved for battery-electric vehicles (BEVs) costing below €47 000 and weighing less than 2.4 t. Households earning under the 5th income decile received a €7 000 rebate; all others were eligible for €5 000. In practice, the overwhelming majority of recipients of the incentives get the €5 000 bonus with only about 10% of households getting the €7 000 (Montout and Robinet, 2024). No incentives have been available for ICE or hybrid since the 2010s and plugins (PHEV) are excluded from any bonus since 2023.

Life-cycle reform in 2024. Since January 2024, eligibility for France’s ecological bonus (“bonus écologique”) is determined by an *environmental score* (ES) calculated by ADEME, a public agency under the French Ministry for the Ecological Transition.⁵ Manufacturers submit detailed bills of materials, and ADEME converts the data into life-cycle CO₂-equivalent emissions per vehicle. This score assesses the carbon footprint of a vehicle throughout its life-cycle, including production, assembly, and transportation. To qualify for the bonus, vehicles must achieve a minimum score of 60 out of 100.

The ES comprises two major components. The main component relates to upstream carbon footprint while a secondary one is about additional environmental factors.

- **Upstream Carbon Footprint ($\geq 70\%$)** includes emissions from:
 - *Ferrous metals*: Carbon footprint from the production of ferrous metals used in the vehicle (excluding the battery).
 - *Aluminum*: Emissions from aluminum production for the vehicle (excluding the battery)—see Figure 1 for the emissions factors used.
 - *Other materials*: Emissions from the production of materials other than ferrous metals and aluminum used in the vehicle.
 - *Battery*: Emissions related to battery production.

⁵ADEME stands for *Agence de la transition écologique*, which translates to French Agency for Ecological Transition. See: <https://www.ademe.fr/> and <https://score-environnemental-bonus.ademe.fr/>.

- *Assembly*: Emissions from intermediate transformations and vehicle assembly, depending on the country of the production site.
- *Transport*: Emissions from transporting the vehicle from the production site to the distribution site in France.

Emission factors by country and mode are shown in Table 1.

- **Additional Environmental Factors ($\leq 30\%$)**: This optional component can contribute up to 30% of the total score and considers:
 - *Use of recycled and bio-sourced materials*: Incorporation of environmentally friendly materials in the vehicle.
 - *Battery reparability*: The ease with which the battery can be repaired or replaced.

While we cannot easily construct this score as we do not have access to all the required quantities per vehicle, we can still precisely determine which cars were eligible and which were not. A ministerial document published on December 14 of 2023 listed car models qualifying for the EV incentive in early 2024. Models whose score exceeded the fixed threshold and were eligible in 2023—roughly 30% of French EV sales in 2023—lost the bonus. The list is updated quarterly.⁶ Our DiD analysis focuses on car models which lost eligibility and compares them to eligible cars as of December 14, 2023. Our descriptive and counterfactual analysis, however, takes into account the expanding boundaries of the set of eligible vehicles over time—which is entirely driven by the introduction of new models and not by cars that were already available experiencing a change in eligibility status. Appendix Figure A.2.1 presents two popular Chinese-made models that lost the bonus—the Dacia Spring and the MG 4.

Figure 1: Illustration of the material factors used in the ES calculation (aluminum)

$FE_{aluminium}^{site}$ sont définies suivant la zone d’implantation du site d’assemblage considéré, selon les dispositions suivantes :

Zone d’implantation du site considéré	Facteur d’émission carbone de la production d’aluminium (pur et allié) par unité de masse aluminium (pur et allié) (kg-eq CO ₂ /kg)
Amérique du Nord	8,5
Amérique du Sud	13,9
Europe	8,6
Chine	20,0
Japon	12,6
Conseil de Coopération du Golfe	11,4
Autre	18,5

⁶For methodology and the current list of eligible models, see: <https://score-environnemental-bonus.ademe.fr/> and the ministerial decree: <https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000048167407>.

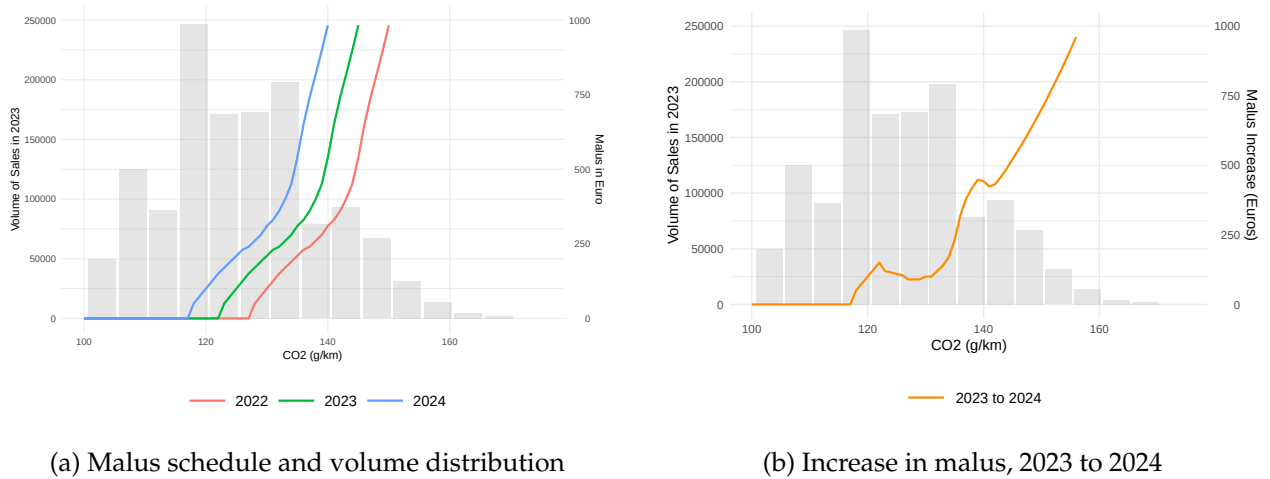
Table 1: Selected emission factors used by ADEME.

Component	France	Germany	China
Ferrous metals (kg CO ₂ -eq./kg)	1.4	1.4	2.0
Aluminum (kg CO ₂ -eq./kg)	8.6	8.6	20.0
Plastics/composites (kg CO ₂ -eq./kg)	4.6	4.6	5.0
Battery (kg CO ₂ -eq./kWh)	53	53	68
Assembly (kg CO ₂ -eq./kg vehicle)	0.58	0.83	1.60
Maritime transport (kg CO ₂ -eq./t km)	0.101 (global)		
Rail transport (kg CO ₂ -eq./t km)	0.010	0.023	0.041
Road transport (kg CO ₂ -eq./t km)	0.208	0.256	0.377

2.1.2 The French feebate

The “malus” tax. France operates a tax on high tank-to-wheel CO₂-rated vehicles, including plug-in hybrids, which rises with emissions. The top bracket reaches €60,000 in 2024. Figure 2 (panel a) documents the schedule focusing on the lower end of the CO₂ distribution where most of the sales occur, and panel (b) shows the jump between 2023 and 2024.

Figure 2: The CO₂ malus: schedule and 2024 increase.



Note: Panel (a) shows the malus tax schedule by CO₂ rating (g/km) for 2022–2024, with the distribution of 2023 sales overlaid. Panel (b) shows the increase in the malus between 2023 and 2024 for each CO₂ level, weighted by 2023 sales volume. The malus applies to all ICE and plug-in hybrid vehicles; BEVs are exempt.

2.1.3 Contemporary policy changes

Social leasing. A temporary “leasing social” scheme subsidised long-term EV leases for low-income households. 50 000 contracts—roughly 25% of all 2023 bonus claims—were signed between January and March 2024 and the program was stopped afterward. Models eligible for the leasing social are a subset of those eligible for the main EV incentive. The cost of the program is €13 000 per vehicle. The program does not impact our estimation of the average treatment effect of losing

eligibility to the EV incentive program. It matters however when we translate these effects into semi-elasticities. We take the program into account including it into the expected increase in post-tax price induced by the loss of eligibility, as detailed in section 2.1.4 (for further details see the French Senate report⁷.) We cannot directly observe which purchase was subsidized by the social leasing. We know however that past March 2024 no new leasing contracts were offered, and the program should therefore not directly impact market shares past this date.

Corporate buyers. From February 2024 onwards, corporate fleets ceased to receive the €3,000 bonus. We weight the bonus change by the 2023 split between household and corporate registrations (80% vs. 20%).

2.1.4 How large is eligibility-loss shock for EV models?

How large is the price shock for a vehicle that loses the EV purchase subsidy? While this exact magnitude has little bearing on our difference-in-differences estimates of average treatment effects, it is crucial for translating those effects into semi-elasticities and for calibrating our counterfactual simulations. We propose to bound the subsidy loss faced by models that fail the 2024 environmental-score (ES) filter. Those bounds are expected values of the loss depending on assumptions regarding the treatment of corporate fleets and the social leasing:

1. Impacted vehicles would not have been eligible to social-leasing programme (lower bound).

In 2023, corporate fleets account for a share $\pi_c = 0.2$ of EV purchases, and household for the remainder $\pi_h = 0.8$ (Montout and Robinet, 2024). Corporate purchases receive a €3,000 bonus, while households receive €5,000 (90% of cases) or €7,000 (10%, with low enough income).⁸ Setting the post-reform subsidy to zero gives

$$\Delta\tau_{\text{low}} = \pi_c(e3\,000/6) + \pi_h[0.9 \times e5\,000 + 0.1 \times e7\,000] \approx e4\,260. \quad (1)$$

This version of the policy shock allows us to simulate a counterfactual policy where, even in the absence of the ES reform, the impacted models would *not* have been eligible for the social leasing.

2. Including social leasing (upper bound). The 2024 “leasing social” scheme signed about 50,000 contracts before being frozen in March.⁹ Each lease embeds an implicit subsidy of roughly €13 000.¹⁰ Removing these vehicles from the post-March data and recalculating the weighted average raises the effective bonus loss to

$$\Delta\tau_{\text{high}} = \pi_c(e3\,000/6) + \pi_h\left\{0.75[0.9 \times e5\,000 + 0.1 \times e7\,000] + 0.25 \times 13\,000\right\} \approx e5\,900 \quad (2)$$

⁷<https://www.senat.fr/rap/a24-148-2/a24-148-212.html>

⁸Based on ASP statistics presented in (Montout and Robinet, 2024).

⁹For comparison, that would be about 25 % of 2023 bonus claims. Budget report, French Senate, <https://www.senat.fr/rap/a24-148-2/a24-148-212.html>.

¹⁰Same source as footnote 9.

The fraction of low-income households who could benefit from the €7,000 EV incentives is set at 10% based on figures reported in Montout and Robinet (2024), consistent with the share used in equation (1). This version of the shock can be used to get the counterfactual situation where ES is not implemented and ES-impacted vehicles would have been eligible to the social leasing.

We therefore treat $\Delta\tau_{\text{low}} = e4.26k$ and $\Delta\tau_{\text{high}} = e5.90k$ as reasonable lower and upper bounds on the policy-induced price increase for disqualified EV models. More specifically, we consider the upper bound of the shock when estimating semi-elasticities empirically as this seems like the best approximation of the price shock faced by potential consumers. For our baseline simulation, however, we try to isolate the impact of the ES reform and consider the lower bound version of the shock—a scenario where the social leasing program would have excluded the ES impacted models anyway.

2.2 Data

Eligibility and sales in France. The Ministry publishes an exhaustive list of models that satisfy the 2024 life-cycle “environmental score” even though the full underlying calculations are not public. We merge that list with ADEME’s monthly registration data at the CNIT level (as in Kessler et al. (2023)) using TVV-CNIT correspondence. CNIT refers to “Code National d’Identification du Type” (national vehicle type identifier), which is the most granular vehicle identifier available. For example, a Renault Zoe with horsepower of 85 KW has a different CNIT identifier than a Renault Zoe with 100 KW. TVV on the other hand refers to “Type Variante Version” (type variant version), a vehicle identifier determined by the producer. We aggregate the data somewhat: a single observation is defined by the tuple (*model, brand, fuel-type, fiscal-power class, curb-weight, month*), which gives us a vehicle-level panel of monthly sales.

Place of assembly and battery origin for EVs. Each vehicle is matched to its assembly plant using S&P global production database. For electric vehicles, we additionally attach the country or region where the battery cells were produced—see Appendix section A.1 for a full description of the process to determine battery country of origin. Combining these locations with weight-based emission factors from the IEA allows us to impute manufacturing CO₂ for every model.

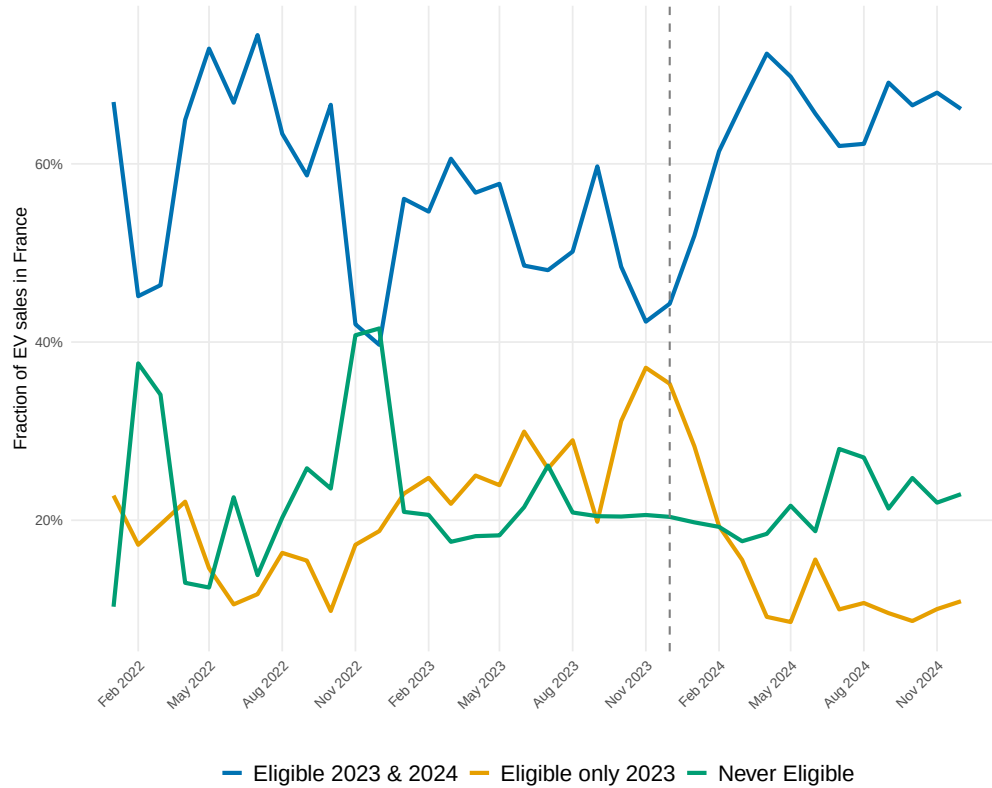
Retention rate moments from an original second-choice survey. Price sensitivity parameters as well as some of the nesting parameters (for ICE) are estimated using policy variation. In order to estimate a subset of the nesting parameters, we use moments from a French stated-preference survey conducted in May 2026 ($N = 5,645$ respondents). Specifically, the share of EV buyers who would still choose an EV if their preferred model were unavailable, and the same statistic conditional on remaining in the same segment (see Appendix B.4 for details and the full substitution heatmaps).

3 Descriptive evidence

Before moving to causal estimation, we present patterns of sales around the date of the reform. We begin by showing the evolution of EV market shares according to the eligibility status, place of assembly, and battery-cell origin (Figures 3–4). We then compare aggregate EV adoption in France and Germany around the reform date (Figure 5). These stylized facts motivate our reduced-form and structural analyses in subsequent sections.

EV market shares by subsidy-eligibility status. Figure 3 shows a pronounced break at the January-2024 reform. Models that were eligible for the bonus in 2023 but lost eligibility thereafter see their share shrink by roughly 60%, and the drop persists through the end of the sample. By contrast, the share of *never-eligible* models remains essentially flat.

Figure 3: Market shares *within* the EV segment by subsidy-eligibility status, 2023–2024

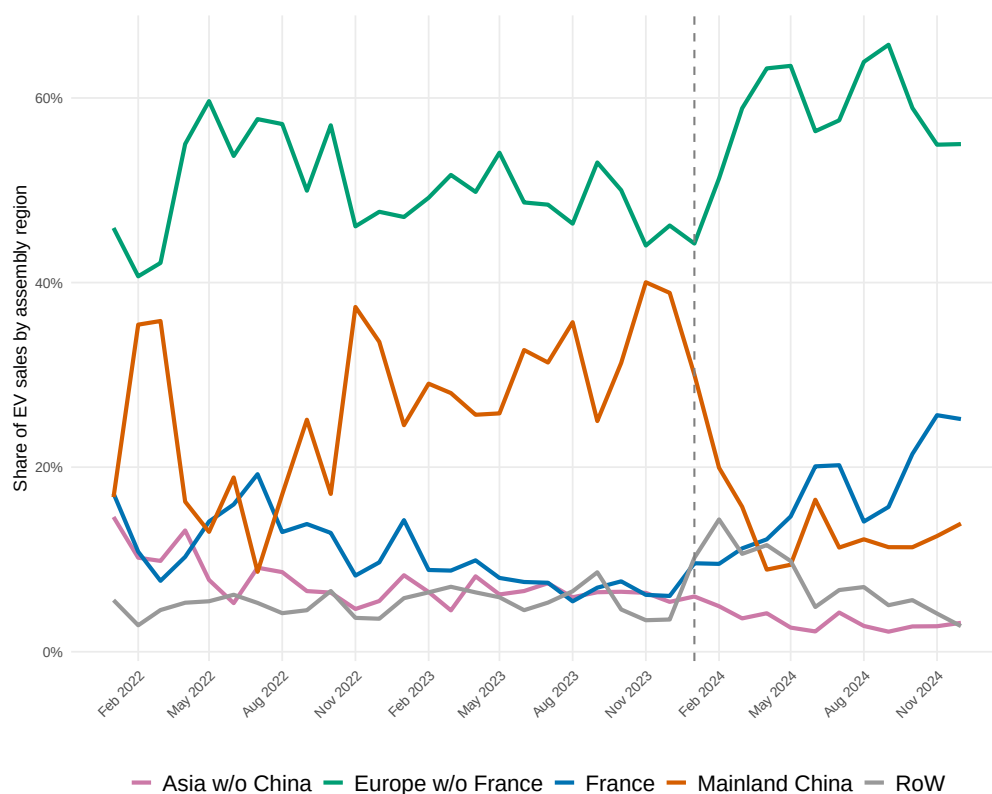


Note: Monthly market shares within the EV segment by subsidy-eligibility status. “Eligible 23 & 24”: models that retained the bonus after the January 2024 environmental-score reform. “Eligible only 23”: models that lost eligibility due to high manufacturing-phase emissions. “Never Eligible”: models that never qualified (typically priced above €47 000). Shares are computed as the number of registrations of each group divided by total EV registrations in that month. The vertical dashed line marks the reform date (January 2024). Source: ADEME/SDS registration microdata.

EV market shares by country of final assembly. Chinese-assembled EVs had been gaining market share in France throughout 2023, but Figure 4 reveals that the life-cycle reform abruptly reverses

that trend. European—especially French—assembly benefits most, while “Rest of World” models remain roughly constant.

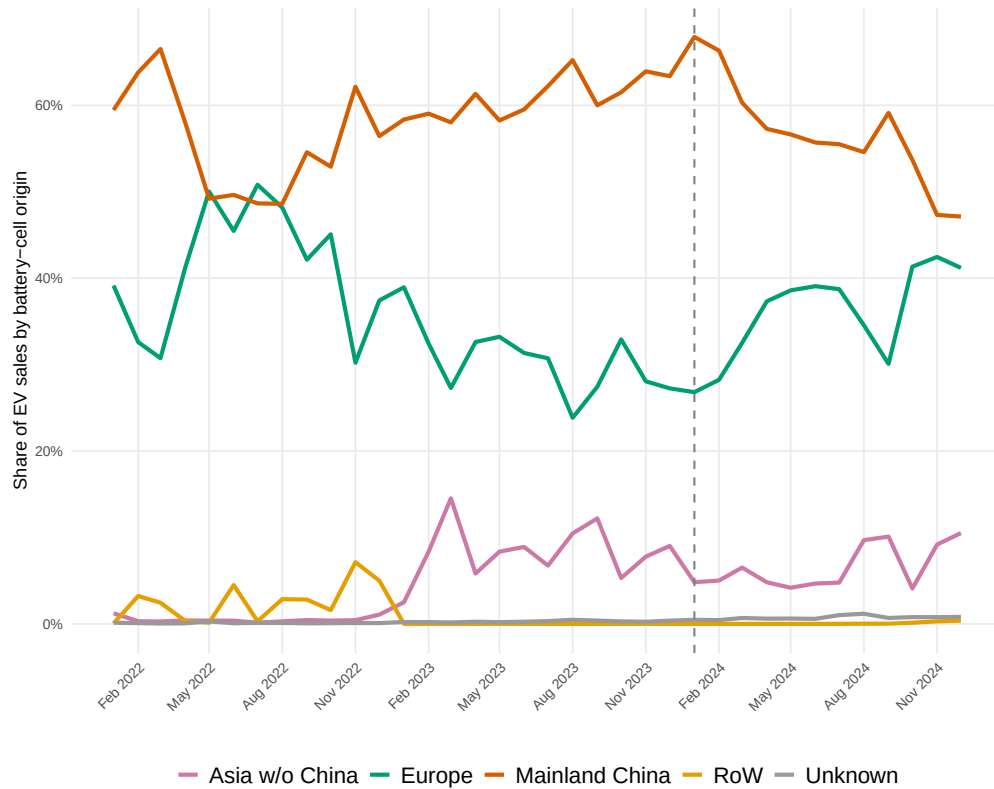
Figure 4: Market shares *within* the EV segment by country (or region) of vehicle assembly



Note: Monthly market shares within the EV segment by country (or region) of final vehicle assembly. Each new vehicle registration is matched to its assembly plant using the S&P Global production database. Shares are computed as registrations assembled in each region divided by total EV registrations in that month. “Europe” excludes France, which is shown separately. For 2025 data, assembly locations are assumed unchanged from end-2024. The vertical dashed line marks January 2024. Source: ADEME/SDES registration microdata matched to S&P Global.

EV market shares by country of battery-cell origin. Battery sourcing is even more concentrated than final assembly: China accounts for about half of all cells sold in 2023. Figure 5 indicates that the eco-score reform first flattens and then marginally reverses the upward trajectory.

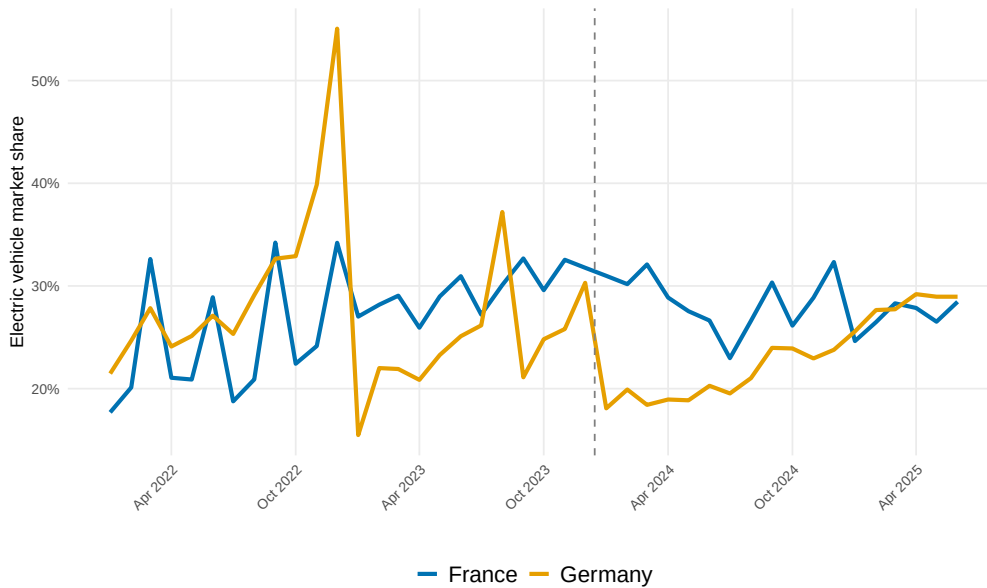
Figure 5: Market shares *within* the EV segment by country (or region) of battery-cell production



Note: Monthly market shares within the EV segment by country (or region) of battery-cell production. Each EV registration is matched to its battery supplier's production location using public and proprietary sources (see Appendix A.1 for details). Shares are computed as registrations with battery cells from each region divided by total EV registrations in that month. "Unknown" includes models for which the battery origin could not be determined. The vertical dashed line marks January 2024. Source: ADEME/SDS registration microdata matched to battery origin data.

Overall EV adoption in France and Germany. While France introduced its life-cycle-based eligibility test for its EV bonus, Germany simultaneously *abolished* its purchase subsidy (both in January 2024). Registrations react sharply in Germany. The share of EVs among new cars falls by roughly one-third (figure 6). In contrast, the French decline is milder and more gradual although we do see visually a slow-down in EV adoption. A synthetic-control comparison with other EU markets confirms a significant break for Germany but finds no statistically clear break for France (see results in the online appendix section B.1), although such cross-country exercise necessarily lacks statistical power and relies on strong identifying assumption. We view this German/French comparison as a motivation for using a structural model as it shows that changes in EV incentives can impact the EV share in the overall market for new passenger cars but that reduced-form approach will not be enough to isolate and estimate precisely such effects when the reforms are not as dramatic as the German case.

Figure 6: Monthly EV market shares in France and Germany.



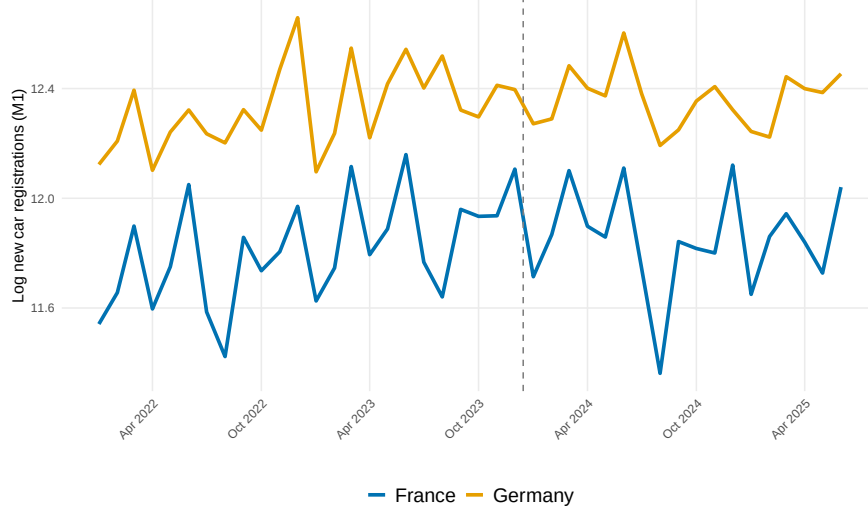
Note: Data from the EU Alternative Fuels Observatory, battery electric vehicle share among newly registered passenger cars (M1).

Overall, the 2024 reform triggered a pronounced reallocation of sales within the French EV segment—towards models that remain eligible to the subsidy—yet the aggregate EV share of total new-car registrations also turns downwards around the same date. Simple synthetic controls give an inconclusive counterfactual for France. To deal with this limitation of data-driven approach to assess overall electrification, we will rely on the structural model developed below to isolate how much of that slowdown is genuinely policy-induced.

Figure 7 displays *total* new-car registrations in France and Germany. We see that they track each other closely throughout the sample period, with strong seasonality but no visible break at the reform date in either country. This is especially informative for Germany, where the complete elimination of EV subsidies in December 2023 caused a dramatic collapse in EV market share (from over 30% to 10%, see Figure 6) yet left aggregate car sales essentially unchanged. If even Ger-

many's far more drastic policy change did not affect total car sales, the milder French LCA reform is unlikely to have done so either. This result motivates the assumption in our subsequent analysis, where we will treat the total market size as given.

Figure 7: Log total new car registrations in France and Germany.



Note: Monthly log total new-car registrations (all powertrains) for France and Germany, 2022–mid 2024. Total registrations are computed as BEV registrations divided by BEV market share. The vertical dashed line marks December 2023, when France introduced the environmental-score reform and Germany eliminated its EV purchase subsidy. Source: EU Alternative Fuels Observatory.

4 Reduced-form empirical results

Having documented the raw shifts in EV shares, we next estimate a difference-in-differences specification that exploits the abrupt withdrawal of bonuses for high-LCA models to identify semi-elasticities of demand for EVs in Section 4.1. Section 4.2 presents the evidence regarding the demand for ICEs, exploiting changes in the malus.

4.1 Evidence on the reform of the EV incentive

Aggregation and econometric specification. For our analysis, we consider monthly sales grouped at the level of the *carline*, which we define as a unique combination of brand, model, and “Puissance Fiscale” (a tax-related binned measure of power).¹¹ This aggregation captures the variation of different trims within a car model, while reducing measurement error that might arise from changes in identifiers (“cnit”) in the raw dataset over time. Appendix section B.2 presents robustness checks for different levels of aggregation. We further restrict the sample to a six months window before and after the policy change. Finally, we only consider models that had positive sales in at least one of the six months prior to the policy change.

¹¹This aggregates the data from the narrow definition of the “cnit”.

A model is labeled as treated if it had a purchase price below 47.000 euro and a weight under 2.4 tons as of December 2023 (i.e. eligible before the reform) and if it lost its eligibility in January 2024. Overall, those represented 31% of our sample of 2023 sales. Car models that were already non-eligible in December 2023 represented around 23% and those that were eligible and retained their status throughout January 2024 made up the remaining 46%. In other words, the reform impacted about a third of the market for electric vehicles in France.

Table 2 presents further summary statistics for the electric vehicle (EV) analysis sample. Summary statistics are presented both for all electric vehicles in the sample, as well as for the subset of those that were eligible in December 2023.

Table 2: Summary Statistics of EV Analysis Sample

Variable	Mean	Std. Dev.	P5	P95	Obs
All models					
Number of Unique Groups	177.0	NA	NA	NA	NA
Number of Unique Treated Groups	36.0	NA	NA	NA	NA
Sales (units)	161.2	487.2	0.0	773.1	2124
\% of 0 market share observations	19.3	NA	NA	NA	NA
Price (€)	42152.3	13725.7	32218.4	130723.3	2124
Ever Eligible models					
Number of Unique Groups	70.0	NA	NA	NA	NA
Number of Unique Treated Groups	36.0	NA	NA	NA	NA
Sales (units)	327.8	727.8	0.0	1642.2	840
\% of 0 market share observations	16.8	NA	NA	NA	NA
Price (€)	37410.0	7156.5	28207.7	46832.3	840

Note: This table reports summary statistics for the EV analysis sample, sub model by month panel. Prices are weighted by sales. Sales are measured in units; prices are filled averages over the sample period.

We use PPML regression which has become the standard theory-consistent estimator for gravity and firm-level market share regressions (on top of accounting for issues related to 0-shares (Silva and Tenreyro, 2006; Chen and Roth, 2024), this ensures the consistency of the fixed effects with the denominator of the market share). We present both a dynamic (3) and a static (4) specification.

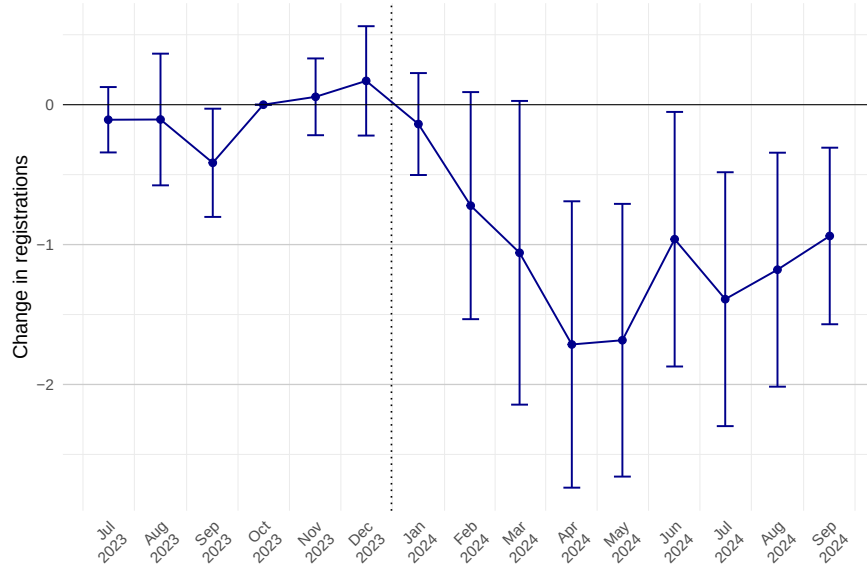
$$\textbf{Dynamic: } s_{j|g,t} = \exp \left[\sum_{k \neq -2} \beta_k \times \mathbb{1}\{k = t\} \times \text{treated}_j + \underbrace{\delta_{s,g,t}}_{\text{seg} \times \text{fuel type} \times t \text{ FE}} + \underbrace{\gamma_j}_{\text{carline FE}} + \epsilon_{j,t} \right], \quad (3)$$

$$\textbf{Static: } s_{j|g,t} = \exp \left[\beta \times \mathbb{1}\{t > 0\} \times \Delta \tau_j + \underbrace{\delta_{s,g,t}}_{\text{seg} \times \text{fuel type} \times t \text{ FE}} + \underbrace{\gamma_j}_{\text{vehicle FE}} + \epsilon_{j,t} \right], \quad (4)$$

where $s_{j|g,t}$ is the share of vehicle j within fuel type $g = \{\text{EV}, \text{ICE}\}$ at time t . Each carline j belongs to a segment s (Sedan, SUV, Minivan, etc.). The dynamic specification is an event study with treatment interacted with months-since treatment dummies. The static regression measures the impact of the amount of subsidy lost ($\Delta \tau_j$), with the lower bound defined in (1). In section 5.4, we discuss the

link between these double difference estimates and the demand parameters of the model we use for counterfactuals, depending on the structure of fixed effects included in the specification.

Figure 8: Event study of loss of eligibility



Note: This figure presents results of a dynamic difference-in-differences regression of the carline's market share in new vehicle registrations on the loss of eligibility according to equation (3). Vertical bars indicate 95% confidence intervals. Specifications with alternative fixed effects can be found in appendix B.2.1. The figures in Appendix B.2.2 present the same analysis for the disaggregated sample.

Diff-in-diffs under interference: what is the causal estimand? In a discrete-choice demand system, the stable unit treatment value assumption (SUTVA) fails: when treated models lose their subsidy, consumers substitute toward control models, inflating their market shares. We briefly characterize the causal estimand of the within-cell DiD under this form of interference, without relying on any functional form for demand.

Denote by $Y_{t,j}(d_j, \kappa_j(\mathbf{D}))$ the potential log-share of model j at time t , where $d_j \in \{0, 1\}$ is j 's own treatment status and $\kappa_j(\mathbf{D})$ summarizes the spillover from the full treatment vector $\mathbf{D}$ onto j 's competitive environment (e.g. how the treatment of other models reshuffles demand toward or away from j). At $t = 0$ (pre-reform), no model is treated and $\kappa_j = 0$, so $Y_{0,j} = Y_{0,j}(0, 0)$. Define the *total* average treatment effect on the treated within fuel type g :

$$ATT_g^T = \mathbb{E}[Y_{1,j}(1, \kappa_j(\mathbf{D}^*)) - Y_{1,j}(0, 0) \mid j \in \mathcal{T}_g], \quad (5)$$

which compares treated models' actual log-shares to a world with no policy at all.

We impose a standard parallel trends assumption: in the absence of any treatment, treated and control models within the same (segment $\times$ fuel type) cell c would have experienced the same

trend in log-shares:

$$\mathbb{E}[Y_{1,j}(0,0) - Y_{0,j}(0,0) \mid j \in \mathcal{T}_c, c] = \mathbb{E}[Y_{1,j}(0,0) - Y_{0,j}(0,0) \mid j \in \mathcal{C}_c, c]. \quad (6)$$

A standard add-and-subtract argument (inserting $Y_{1,j}(0,0)$ as the counterfactual for each group and using (6) to cancel the trend terms) shows that the within-cell DiD identifies:

$$\mathbb{E}[\widehat{\text{DiD}}_g] = \text{ATT}_g^T - \underbrace{\mathbb{E}[Y_{1,j}(0, \kappa_j(\mathbf{D}^*)) - Y_{1,j}(0,0) \mid j \in \mathcal{C}_g]}_{\text{Spillover}_{\mathcal{C}_g} > 0}. \quad (7)$$

The second term, $\text{Spillover}_{\mathcal{C}_g}$, captures the demand gained by control models through substitution from treated models—it is positive in any demand system with downward-sloping demand. Since $\text{ATT}_g^T < 0$ and $\text{Spillover}_{\mathcal{C}_g} > 0$, the DiD overstates the total treatment effect in absolute value: $|\widehat{\text{DiD}}_g| > |\text{ATT}_g^T|$.

Butts (2021) develops a similar decomposition in a spatial setting and proposes selecting comparison groups that minimize spillovers. In our demand context, however, such a strategy is impractical: models that satisfy parallel trends (similar products in the same cell) are precisely those most exposed to substitution spillovers. We therefore do not attempt to eliminate $\text{Spillover}_{\mathcal{C}_g}$ from the DiD. Instead, we estimate the composite $\widehat{\text{DiD}}_g$ and use the demand model to decompose it.

The demand model introduced in Section 5 will provide the structure needed to decompose $\widehat{\text{DiD}}_g$ into ATT_g^T and $\text{Spillover}_{\mathcal{C}_g}$.

Results: DiD for EVs. Figure 8 presents the event study graph from equation (3) where treatment is a dummy indicating eligibility loss. In March 2024, we estimate a coefficient around -1 , which indicates a decrease in sales of about 60% to be compared to October 2023, our reference point. We see very little anticipation in November and December 2023, confirming that the policy change was quite unexpected. Sales of newly ineligible carlines fall further in the next months before stabilizing. Appendix Figures B.2.1 present the same analysis with different sets of fixed effects. Appendix figures B.2.2 present the same analysis for the disaggregated sample. In appendix B.2.1, we use an alternative specification with firm (e.g. Volkswagen, Stellantis, General Motors, etc.) by segment by month fixed effects.

Table 3 presents the static results of equation (4) for all EVs in the sample across multiple sets of fixed effects. The coefficients capture the effect of a one thousand euro price increase through loss of eligibility. The coefficients are negative and significant across the specifications, indicating that losing eligibility decreased the market share of the concerned car models. A robustness check where sales are aggregated at the most granular level (“cnit” instead of carline) is presented in appendix Table B.2.1. As will be made clear in section 5.4.2, the choice of fixed effects of columns (2) and (3) allow for a clear mapping between our estimate and demand parameters of the model. To get a sense of the magnitude, the preferred coefficient of about -0.15 (column 2) implies that a one-thousand-euro increase in the effective price reduces within-segment EV market share by roughly 14%. For the average treated model, which lost a bonus worth approximately 5 KEUR,

this translates into a sales decline of about 55–60%, consistent with the event-study estimates of Figure 8. Under a simple logit benchmark, this semi-elasticity implies an own-price elasticity of about -4.2 , which is in the range of estimates from the existing literature on automobile demand.¹²

Table 3: The impact of the loss of eligibility on new vehicle registrations

Dependent Variable: Model:	Market Share		
	(1)	(2)	(3)
<i>Variables</i>			
Treatment (TP)	-0.130*** (0.045)	-0.156*** (0.042)	-0.215*** (0.073)
<i>Fixed-effects</i>			
Carline	Yes	Yes	Yes
Month	Yes		
Month $\times$ Segment		Yes	
Month $\times$ Segment $\times$ Firm			Yes
<i>Fit statistics</i>			
Observations	2,124	2,124	1,898
Pseudo R ²	0.230	0.234	0.245
Implied own-price elasticity	-3.53	-4.25	-5.84

Note: This table presents results of a static difference-in-differences regression of number of new vehicle registrations on loss of eligibility according to equation (4). Coefficients capture the effect of a 1000 euro price increase. The time period considered covers six pre- and six post-reform months. Observations are grouped at the carline (brand $\times$ model $\times$ fiscal power) level. The same regression with observations grouped at the cnit level can be found in appendix table B.2.1. Appendix tables B.3.1 presents results for a regression controlling for estimated markups. Clustered (carline) standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

4.2 Evaluation of the impact of the feebate

Aggregation and econometric specification. For the analysis of the change in the malus, from December 2023 to January 2024, we consider observations grouped at the brand, model, and CO₂ (g/km) level. Taking into account CO₂ emissions is necessary insofar as it is the primary variable determining the level of the malus. As with EVs, we restrict the sample to a six months window before and after the policy change, where we impose that at least one of the six pre-policy change months must have displayed positive sales. A final restriction that we impose is to exclude vehicles experiencing a malus increase of more than 20,000 euros. While those only represent a small fraction of sales, they are the most polluting luxury cars, which might have fundamentally different demand responses to price changes.

¹²For instance, Goldberg and Verboven (2001) report own-price elasticities between -4.1 and -6.2 for European countries in the 1990s (with -5.6 for France), Grieco et al. (2024) estimate -5.4 for the US market in 2015, and Conlon and Gortmaker (2020) report -3.9 using BLP on US data from the 1970s–1980s.

Table 4 presents summary statistics. It shows that the ICE analysis sample, which includes all vehicles but electric ones, has a somewhat finer grouping than the preferred aggregation of EVs, which is due to the much more differentiated supply of ICE models.

Table 4: Summary Statistics of ICE Analysis Sample

Variable	Mean	Std. Dev.	P5	P95	Obs
Diesel & Petrol models					
Number of Unique Groups	1994.0	NA	NA	NA	NA
Sales in units per month	35.4	156.8	0	166	23927
\% of 0 market share observations	53.8	NA	NA	NA	NA
Price (je2¿82¿ac¿)	27545.3	14352.6	20250	144150	23783

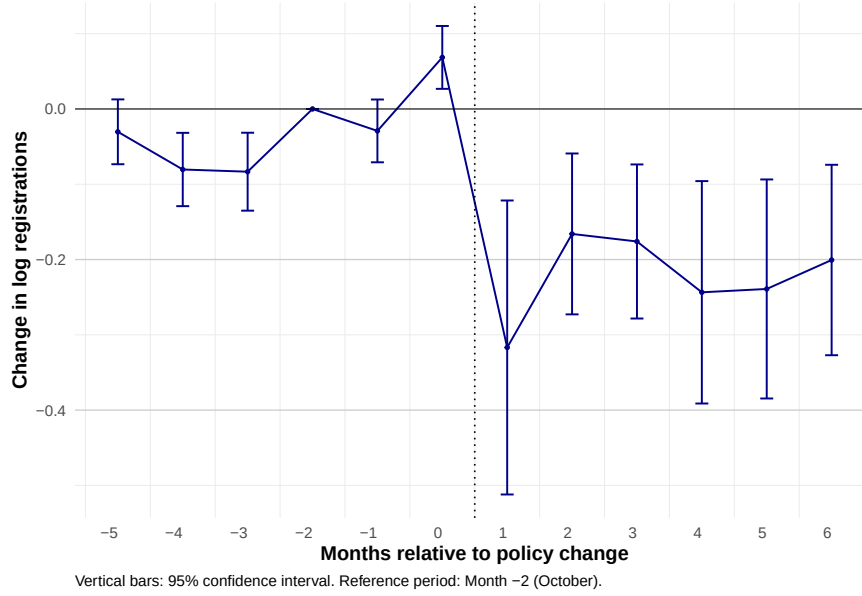
Note: This table reports summary statistics for the ICE analysis sample, sub model by month panel. Prices are weighted by sales. Registrations are measured in units; prices are filled averages over the sample period.

Results: DiD for ICEs. Figure 9 presents the event study graph using a stacked design that pools the 2022–2023 and 2023–2024 malus increases, with treatment measured as the continuous amount of the malus change (in thousands of euros). The specification stacks both reform episodes, providing additional statistical power relative to using only the 2023–2024 change. The coefficients capture the semi-elasticity of within-segment market share with respect to the malus increase. The figure shows a clear negative effect emerging at the reform date, with no evidence of differential pre-trends. A binary-treatment version of this event study is presented in Appendix Figure B.3.1.

Table 5 shows the benchmark results for the effect of the 2023 to 2024 increase in malus on market share. Coefficients capture the effect of a 1000 euro price increase. As in the analysis of the bonus, they are negative and significant across the three sets of fixed effects. The preferred coefficient of about -0.17 (column 2) implies that a one-thousand-euro malus increase reduces within-segment ICE market share by roughly 15%. This semi-elasticity is strikingly close to the one estimated for EVs (-0.15), despite coming from an entirely independent source of variation—the increase of a CO₂-based tax rather than the withdrawal of a purchase bonus—applied to a different vehicle population. The implied own-price elasticity under a simple logit is about -5.0 , again consistent with existing estimates for mature car markets.¹³ As discussed in section 5.4.2, both sets of results will allow us to identify important parameters of the demand system we use to build counterfactual scenarios.

¹³See also Klier and Linn (2015) and Yan and Eskeland (2018) for evidence on the responsiveness of ICE demand to CO₂-based taxation in France, Germany, Sweden, and Norway.

Figure 9: Event study of increase in malus among ICE (stacked, continuous treatment)



Note: This figure presents results of a dynamic difference-in-differences regression of within-segment market share on the continuous malus increase (in K EUR), stacking the 2022–2023 and 2023–2024 reform episodes according to equation (3). Vertical bars indicate 95% confidence intervals. Observations are grouped at the Brand $\times$ Model $\times$ CO₂ g/km level. The regression includes group and segment $\times$ month fixed effects.

Table 5: The impact of the malus increase on new vehicle registrations (Diesel and petrol cars)

Dependent Variable:	Market Share		
Model:	(1)	(2)	(3)
<i>Variables</i>			
$\Delta\tau_j \times Post$	-0.191*** (0.060)	-0.178*** (0.042)	-0.212*** (0.061)
<i>Fixed-effects</i>			
Carline	Yes	Yes	Yes
Month $\times$ Motorisation	Yes		
Month $\times$ Segment $\times$ Motorisation		Yes	
Month $\times$ Segment $\times$ Firm $\times$ Motorisation			Yes
<i>Fit statistics</i>			
Observations	23,927	23,717	23,058
Pseudo R ²	0.230	0.230	0.230
Implied own-price elasticity	-5.51	-5.14	-6.13

Note: This table presents results of a static difference-in-differences regression of number of new vehicle registrations on malus increase in 1000 of euros according to equation (4). The time period considered covers six pre- and six post-reform months. Observations are grouped at the brand $\times$ Model $\times$ CO₂ g/km. Appendix tables B.3.1 presents results for a regression controlling for estimated markups including all ICE vehicles (petrol, gasoline, plugin, hybrid). Clustered (carline) standard-errors in parentheses. Signif. Codes: ***, 0.01, **, 0.05, *, 0.1.

5 Conceptual framework

Providing a complete evaluation of the policy requires not only understanding how responsive demand for impacted vehicles is, but also towards which kind of alternatives the suppressed demand was reallocated. To do so, we need to recover the substitution patterns across models, which are not directly observable from the reduced-form regressions. In this section, we embed our semi-elasticity estimates into a three-level nested-logit framework. Section 5.1 lays out the model and section 5.2 presents the associated choice probabilities. Section 5.3 specifies the mean utility associated with each vehicle and presents our approach to identification. In particular, section 5.4 presents our sequential approach to identification. We show how second-choice survey moments pin down the nest parameters (5.4.1) and then describe how appropriately parametrized versions of our DiD regressions identify a combination of the remaining parameters (5.4.2). We present the results in section 5.5.

5.1 Nested-logit structure

We consider a static model of car choice in the spirit of Goldberg (1995) and following a long tradition of discrete choice models (McFadden et al., 1978; Berry, 1994).

Passenger-car demand is formalized as a three-level nested logit. Consumers first choose between buying a given power-train $g \in \{\text{EV}, \text{ICE}\}$ or not buying a car; within each g they then pick a *segment* s (*SUV*, *Sedan*, *Minivan*, *Break*, *Sports*, *Other*); finally they choose a precise carline j (defined by a brand, model, and power rating combination). Figure 10 illustrates the decision tree, whose *branch parameters* $\mu_{4,g}$ (within-segment) and μ_3 (across segments) regulate substitution patterns.¹⁴ Equivalently, the nested logit can be seen as imposing a block-diagonal structure on the correlation matrix of the idiosyncratic errors across products: two models in the same (fuel type, segment) cell have correlation $1 - (\mu_{4,g}/\mu_3)^2$; two models in the same fuel type but different segments have correlation $1 - \mu_3^2$; and models in different fuel types are uncorrelated. The constraints $0 < \mu_{4,g} \leq \mu_3 \leq 1$ ensure a valid (positive-definite) correlation matrix. Appendix Table C.1.1 displays this structure explicitly.

The ordering of the nesting levels—fuel type first, then segment—is supported by the data. Appendix C.2 provides the hat-algebra derivations of the retention rates under both orderings and shows that only the fuel-type-first nesting can jointly match the empirical retention rates from second-choice data.

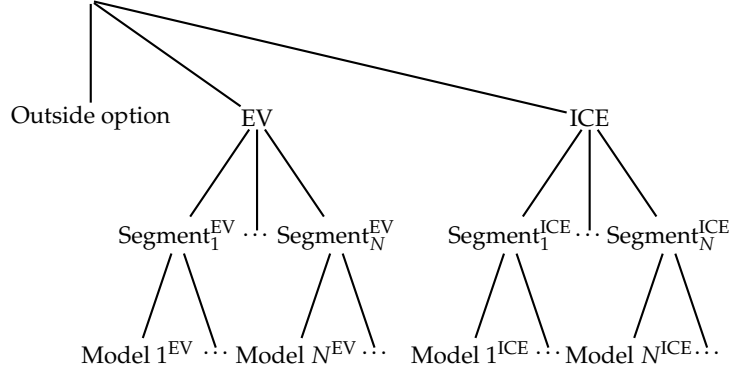
5.2 Choice probabilities

Denote by s_j the market share or choice probability for carline j among *all* options and by s_g° , $s_{s|g}^\circ$, and $s_{j|s,g}$ the conditional shares at each stage.¹⁵ The nested-logit algebra delivers the three-way factorization

¹⁴When $\mu_{4,g} = \mu_3 = 1$ the nests collapse to the standard logit. Smaller values increase within-nest substitution.

¹⁵The subscript $\circ$ is used to signal market shares aggregated at a higher level of aggregate than the carline (segment or fuel-type).

Figure 10: Decision tree in the two-level nested logit.



Note: Consumers first choose a fuel type $g \in \{EV, ICE\}$, then a segment s (e.g. sedan, SUV), and finally a specific vehicle j . The nesting parameters μ_3 and $\mu_{4,g}$ govern within-nest substitution at the segment and vehicle levels respectively. A smaller μ implies stronger within-nest correlation and more substitution among products sharing the same nest.

We can express the choice probability for carline j as the product of several nested conditional probabilities

$$s_j = s_g^\circ \times s_{s|g}^\circ \times s_{j|s,g} \quad (8)$$

and the nested logit algebra delivers closed forms for each component of this factorization:

$$s_{j|s,g} = \frac{\exp(V_{jt}/\mu_{4,g})}{\sum_{k \in (s,g)} \exp(V_{kt}/\mu_{4,g})}, \quad s_{s|g}^\circ = \frac{\exp(IV_{s|g,t}/\mu_3)}{\sum_{r \subset g} \exp(IV_{r|g,t}/\mu_3)}, \quad \text{and} \quad s_g^\circ = \frac{\exp(IV_{g,t})}{1 + \sum_h \exp(IV_{h,t})} \quad (9)$$

and inclusive values given by

$$IV_{s|g} = \mu_{4,g} \ln \left[\sum_{k \in (s,g)} \exp(V_{kt}/\mu_{4,g}) \right], \quad \text{and} \quad IV_g = \mu_3 \ln \left[\sum_{r \subset g} \exp(IV_{r|g,t}/\mu_3) \right]. \quad (10)$$

5.3 Indirect utility and the estimating equation

We now specify the deterministic component of carline j 's utility in month t :

$$V_{jt} = -\alpha p_{jt} + \xi_j + \zeta_{jt}, \quad (11)$$

where p_{jt} decomposes into the policy instrument (τ_{jt}), marginal cost (mc_j), and a possibly endogenous mark-up ($\tilde{p}_{jt}^0$):

$$p_{jt} = \tau_{jt} + mc_j + \tilde{p}_{jt}^0.$$

It will be useful to define the market share of j within EVs, defined as $s_{j|g,t} \equiv s_{j,t}/s_{g,t}^\circ$. Using (9), (10), and (11) and taking logs, we obtain that

$$\ln s_{j|g,t} = -\frac{\alpha}{\mu_{4,g}}\tau_{jt} + \frac{\zeta_j - \alpha mc_j}{\mu_{4,g}} - \underbrace{\left(\frac{1}{\mu_{4,g}} - \frac{1}{\mu_3}\right) IV_{s|g,t} - \frac{IV_{g,t}}{\mu_3}}_{\text{varies at the } gst \text{ level}} - \frac{\alpha}{\mu_{4,g}}\tilde{p}_{jt}^0 + \frac{\zeta_{jt}}{\mu_{4,g}}. \quad (12)$$

Equation (12) highlights the difficulty of applying standard instrumental variable to identify all parameters of the model. In particular, to identify separately $\mu_{4,EV}$ and μ_3 one would need to instrument for the overall attractiveness of EV at a given time period $IV_{EV,t}$. This is unattractive as it is difficult to isolate time series variation in the relative attractiveness of EV at the aggregate level that is not confounded by other possible shocks (changes in preferences, business cycle, etc). We therefore consider a specification with a rich set of fixed effects which will allow us to recover $\frac{\alpha}{\mu_{4,g}}$. The term in braces depends only on the $\langle s, g, t \rangle$ combination; it is purged with segment-fuel type-month fixed effects δ_{sgt} , whereas the time-invariant $\{\zeta_j - \alpha mc_j\}$ are absorbed by model fixed effects γ_j . We can therefore write equation (12) under a form that matches the key estimation equation (4) which was fitted in section 4:

$$\ln s_{j|g,t} = -\frac{\alpha}{\mu_{4,g}}\tau_{jt} + \delta_{sgt} + \gamma_j + u_{jt}, \quad \text{with} \quad u_{jt} = -\frac{\alpha}{\mu_{4,g}}\tilde{p}_{jt}^0 + \frac{\zeta_{jt}}{\mu_{4,g}}. \quad (13)$$

In section 5.4.2, we discuss how exogenous changes in τ_{jt} (bonus withdrawals for EVs and malus hikes for ICEs) can be leveraged to identify the key semi-elasticities $\alpha/\mu_{4,g}$ under different identifying assumptions. In a second step, we use U.S. second-choice data to pin down the nesting parameters $\{\mu_3, \mu_{4,EV}\}$; together these deliver α and $\mu_{4,ICE}$, completing the demand system.

5.4 Identification and estimation

The nested nature of the model allows for a sequential identification of the parameters of the model. We first show how second choice data pin down the parameters $\{\mu_3, \mu_{4,EV}\}$ and then we show how to relate our DiD estimates to $\alpha/\mu_{4,g}$.

5.4.1 Second choices as a source of identification in the nested logit model

We build on papers using second choice data to estimate a model of discrete choice for differentiated goods (e.g. Berry et al., 2004). We use two empirical moments from a French stated-preference survey (May 2026, $N = 5,645$; see Appendix B.4). Our procedure relies on expressing the retention rates as a function of parameters and observed shares. As such, it bypasses the need to estimate the mean utility V_j associated with each choice j . It also makes explicit the absence of dependence of these parameters and moments on the price coefficient α .

Our *second-choice* survey asked each respondent which vehicle they would have purchased had their first choice been unavailable. Replicating that hypothetical removal in the model provides moments that map directly to the two unknown branch parameters μ_3 (segment level) and $\mu_{4,EV}$ (vehicle level inside the EV nest).

Inclusive values within the EV nest. Let s denote a segment inside the EV branch and V_j the deterministic utility of model j . *Inclusive values* become

$$IV_{s|EV} = \ln \left(\sum_{k \in (s, EV)} e^{V_k / \mu_{4, EV}} \right), \quad IV_{EV} = \ln \left(\sum_{r \in EV} e^{\mu_{4, EV} / \mu_3 IV_{r|EV}} \right). \quad (14)$$

Given (14), the three conditional choice probabilities can be rewritten as

$$s_{j|s, EV} = \frac{e^{V_j / \mu_{4, EV}}}{\sum_{k \in (s, EV)} e^{V_k / \mu_{4, EV}}}, \quad s_{s|EV}^\circ = \frac{e^{\mu_{4, EV} / \mu_3 IV_{s|EV}}}{\sum_{r \in EV} e^{\mu_{4, EV} / \mu_3 IV_{r|EV}}}, \quad \text{and} \quad s_{EV}^\circ = \frac{e^{\mu_3 IV_{EV}}}{1 + \sum_{g \in \{EV, ICE\}} e^{\mu_3 IV_g}}.$$

Removing a single model. Suppose model $j \in (s, EV)$ is withdrawn. At the lower level, the segment's inclusive value falls by $\Delta IV_{s|EV} = \ln(1 - s_{j|s, EV}) < 0$. Propagating this shock upward yields a change in the EV nest's attractiveness:

$$\Delta IV_{EV} = \ln \left(1 - s_{s|EV}^\circ + s_{s|EV}^\circ (1 - s_{j|s, EV})^{\mu_{4, EV} / \mu_3} \right). \quad (15)$$

Because both $s_{j|s, EV}$ and $s_{s|EV}^\circ$ are *observed* pre-removal shares, ΔIV_{EV} is a closed-form function of the unknown pair $(\mu_3, \mu_{4, EV})$.

Retention rate moments. We can express two retention rates predicted by the nested logit structure. The first one is the fraction of j buyers that would keep on buying an EV, should j be removed from the choice set. The second one is the same concept computed at the $EV \times$ segment level.

Using equation (15), the post-removal EV share is $s_{EV}^\circ(\text{no } j) = s_{EV}^\circ \Delta_j / [s_{EV}^\circ \Delta_j + (1 - s_{EV}^\circ)]$, where $\Delta_j = \exp(\mu_3 \Delta IV_{EV})$.

The EV retention rate after removing vehicle j is therefore

$$\eta_j^{(EV)} = \frac{s_{EV}^\circ(\text{no } j) - (s_{EV}^\circ - s_j)}{s_j}, \quad 0 < \eta_j^{(EV)} < 1. \quad (16)$$

The EV/segment retention rate implied by theory is

$$\eta_j^{(s, EV)} = \frac{s_{EV}^\circ(\text{no } j) s_{s|EV}(\text{no } j) - (s_{EV}^\circ s_{s|EV} - s_j)}{s_j}, \quad (17)$$

with $s_{s|EV}(\text{no } j) = \frac{s_{s|EV} \Lambda_j}{s_{s|EV} \Lambda_j + 1 - s_{s|EV}}$ and $\Lambda_j = (1 - s_{j|EV, s})^{\mu_4 / \mu_3}$.

Averaging over all EV vehicles, we obtain:

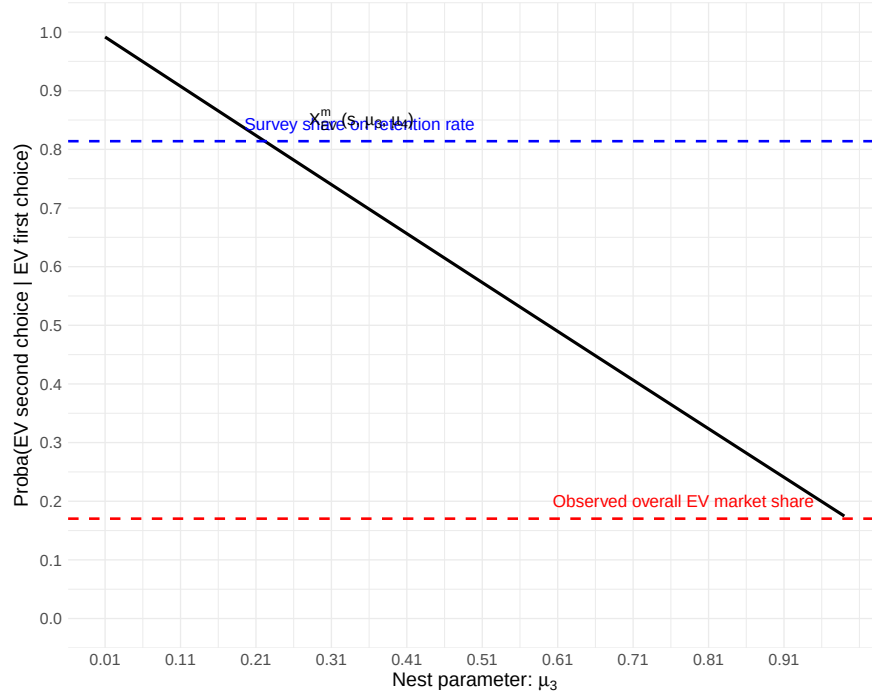
$$X_{EV}^m(\mu_3, \mu_{4, EV}) = \sum_{j \in EV} s_{j|EV} \eta_j^{(EV)}, \quad \text{and} \quad X_{s, EV}^m(\mu_3, \mu_{4, EV}) = \sum_{j \in EV} s_{j|EV} \eta_j^{(s, EV)}. \quad (18)$$

As empirical counterparts to those retention rates, we use the French survey moments (Appendix B.4): $X_{EV}^o = 0.814$ and $X_{s, EV}^o = 0.332$. Matching $X_{EV}^m = X_{EV}^o$ and $X_{s, EV}^m = X_{s, EV}^o$ pins down

the nest scales $\{\mu_3, \mu_{4,EV}\}$ without requiring any information on the attractiveness of each product or the price coefficient α .

Illustration. For illustrative purposes, we draw $X_{EV}^m(\mu_3, \mu_{4,EV})$ in figure 11 under the simplifying assumption $\mu_{4,EV} = \mu_3$: the retention rate declines sharply as μ_3 rises toward the logit limit, illustrating the moment’s identifying power.

Figure 11: Model-predicted EV-on-EV retention rate as a function of μ_3 when $\mu_{4,EV} = \mu_3$.



Note: This graph displays the EV retention rate in a single-level nested logit (ICE/EV) as a function of μ_3 , using French data on observed market share in 2023. The blue line represents the empirical EV retention rate from the French second-choice survey (May 2026): $X_{EV}^o = 0.814$.

Once $\{\mu_3, \mu_{4,EV}\}$ are pinned down, the quasi-experimental estimates of $\alpha/\mu_{4,EV}$ and $\alpha/\mu_{4,ICE}$ from regression (13) deliver the full set of demand parameters $\{\alpha, \mu_3, \mu_{4,EV}, \mu_{4,ICE}\}$, closing the model used in Sections 6.

5.4.2 Mapping the DiD estimates to the model price parameter α

Going back to equation (13), we now discuss whether, and under what assumptions, the regressions presented in section 4.1 identify the ratio $\alpha/\mu_{4,g}$.

(0) Monopolistic competition in pricing. If firms price at marginal cost (i.e. $\tilde{p}_{jt}^0 = 0$ as in the perfectly competitive case) or if they do not vary conditional on firm and time fixed effect (monopolistic competition), the policy variable τ_{jt} is uncorrelated with the unobserved demand shocks ξ_{jt} ; hence equation (13) can be estimated directly, provided segment-fuel type-period (gst) fixed effects

are included. These coefficients are presented in columns (2) of Table 3 and Table 5 for ICE and EVs respectively.

(1) Absorbing mark-ups with firm-gst fixed effects. Under multi-product Bertrand pricing and nested logit demand, mark-ups vary at the firm $\times$ g $\times$ s $\times$ t level—see Appendix C.4.1. Introducing firm $\times$ g $\times$ s $\times$ t fixed effects absorbs those mark-up changes and breaks any spurious correlation between τ_{jt} and $\hat{p}_{jt}^0$. These coefficients are presented in columns (3) of Table 3 and Table 5 for ICE and EVs respectively, and tend to be somewhat larger than the one including only segment-fuel type-period (gst) fixed effects. Whereas this specification is consistent with the theoretical framework (Nested logit demand with Bertrand multiproduct pricing), it throws away a lot of the variation to identify the parameter. Accordingly, we prefer the approach presented below which is theory consistent and preserves more of the variation.

(2) Iterative control for endogenous mark-ups. The parameters $\{\mu_3, \mu_{4,EV}\}$ are identified from second choice data. Starting from an initial guess for the structural parameters $\{\alpha, \mu_{4,ICE}\}$, we compute model-implied mark-ups $m_{jt}(\theta)$ where θ gathers all 4 structural parameters, add $\alpha m_{jt}(\theta) / \mu_{4,g}$ to the dependent variable in equation (13), re-estimate equation (13), update θ with a new guess for $\alpha / \mu_{4,g}$, and repeat until convergence (appendix C.4.2). This procedure delivers a consistent semi-elasticity even when mark-ups respond endogenously to policy (see Head and Mayer, 2021; Breinlich et al., 2024).

Appendix table B.3.1 presents these results.¹⁶ Results are very similar, suggesting predicted changes in markups are small and largely independent from the reform, consistent with the idea that market shares are fairly small and pass-through is close to perfect.¹⁷

5.5 Estimates

The price semi-elasticities that anchor the demand system come from our difference-in-differences design: losing access to the EV purchase bonus shifts sales with an estimated semi-elasticity $\hat{\beta}_{EV} = -0.1564$, while the 2024 increase in the CO₂-malus for ICEs implies $\hat{\beta}_{ICE} = -0.1775$. Both coefficients stem from the specification with segment–fuel type-time and vehicle fixed effects and where we control for possible changes in markups (see Table B.3.1). Combining these coefficients with the second-choice statistics delivers the full set of nested-logit parameters collected in Table 6.¹⁸

¹⁶In a Nash-Bertrand equilibrium, pre-tax prices will adjust to changes in malus, through changes in markups (i.e. there’s imperfect pass-through). Markups are functions of market shares, demand parameters and tax policy. Changes in markups are likely correlated with changes in tax / malus and will impact demand, thus presenting a potential threat to identification in our regression. As presented in Appendix section C.4.2, we deal with this by constructing markups based on observed market shares and assumed demand parameters. We then adjust the market share for this term assuming a given value for the within-nest price sensitivity and run the estimation using the adjusted dependent variable. Based on the results, we update the demand parameters and markups accordingly and iterate until convergence.

¹⁷If anything, they are slightly more negative, suggesting that endogenous changes in markups induce a positive but small bias in our baseline estimate, consistent with the notion that, on average markups have decreased slightly in response to increases in the feebate.

¹⁸Standard errors and bootstrap confidence intervals for $\hat{\mu}_3$ and $\hat{\mu}_{4,EV}$ are reported in Appendix C.3. Inference uses the respondent-level sample covariance of the second-choice indicators (Appendix B.4) together with the delta method, and is cross-checked against a nonparametric bootstrap over respondents.

The upper-nest correlation $\hat{\mu}_3 = 0.267$ indicates substantial within-fuel type substitution relative to across-fuel type substitution, and the lower-nest scale differs markedly between EVs and ICEs ($\hat{\mu}_{4,\text{EV}} = 0.219$ vs. $\hat{\mu}_{4,\text{ICE}} = 0.193$), confirming that ICE buyers view products within a segment as closer substitutes than do EV buyers.

Table 6: Estimated preference parameters

Source	Point estimate	
	Parameter	Value
Diff-in-diffs (bonus & malus)	$\hat{\beta}_{\text{EV}}, \hat{\beta}_{\text{ICE}}$	−0.1564, −0.1775
Second choice data ^a	$\hat{\mu}_3, \hat{\mu}_{4,\text{EV}}$	0.267, 0.219
Combined identification	$\hat{\alpha}, \hat{\mu}_{4,\text{ICE}}$	0.034, 0.193

^a French stated-preference second-choice survey, May 2026 ($N = 5,645$). See Appendix B.4.

To validate the model beyond the moments used for estimation, we compare its predictions for ICE retention behaviour—left entirely untargeted—with their empirical counterparts. As reported in Table 7, the simulated probability that an ICE buyer remains in the ICE nest when her top choice is unavailable is 0.966 (versus 0.954 in the data), a good match at the nest level. The segment-level retention (the probability that an ICE buyer’s second choice is also a non-EV in the same segment) is predicted at 0.510 against an empirical value of 0.385; the model over-predicts within-segment loyalty for ICE buyers, reflecting that the segment parameters were calibrated on EV second-choice patterns where within-nest loyalty is more pronounced. Overall, the model replicates the nest-level substitution patterns reasonably well outside the EV calibration window.

Table 7: Untargeted retention moments (ICE nest)

	Model	Data
X_{ICE}^m : P(non-EV $\rightarrow$ non-EV)	0.967	0.954
$X_{s,\text{ICE}}^m$: P(non-EV $\rightarrow$ non-EV, same seg.)	0.445	0.386

Revisiting the DiD estimand with the demand model. We can now revisit the decomposition of the DiD estimand introduced in Section 4.1. Under the nested logit, the spillover function $\kappa_j(\mathbf{D})$ operates through inclusive values that are common to all models within a (segment $\times$ fuel type) cell. This has two implications. First, the DiD coefficient maps directly to the within-nest semi-elasticity $-\alpha/\mu_{4,g}$, which is a well-defined structural parameter regardless of the spillover. Second, combined with the nesting parameters $(\mu_3, \mu_{4,g})$ estimated from second-choice data, we can quantify the role of spillovers in the continuous treatment setting.

The DiD coefficient $\hat{\beta} = -\alpha/\mu_{4,\text{EV}} = -0.156$ measures the effect of a one-thousand-euro price increase on the within-fuel-type log market share. However, the actual per-euro effect on treated models—computed from the counterfactual log-share change divided by each model’s treatment intensity $\Delta\tau_j$ —is only -0.128 . The difference, $+0.029$ per KEUR, reflects the inclusive-value feed-

back: when treated models lose demand, the attractiveness of the EV nest falls, partially offsetting the within-nest reallocation toward control models. On the control side, untreated EVs gain +0.023 log points per K EUR of average treatment on treated models. Overall, $\hat{\beta}$ overstates the actual per-euro treatment effect by about 18%. The spillover is economically meaningful but modest, confirming that the DiD coefficient is a reliable guide to the magnitude of the demand response.

6 Cost benefit analysis and counterfactuals

We present the method to construct counterfactuals in terms of deviation from observed shares in section 6.1. In section 6.2, we present the results of a counterfactual policy where no environmental score reform took place.¹⁹ In section 6.3, we carry out a cost benefit analysis of the actual reform, relying on additional assumptions, and evaluate its fiscal efficiency through the Average Value of Public Funds (AVPF), accounting for producer surplus and fiscal externalities. We then implement a budget neutral alternative policy where savings are recycled to increase incentives on still eligible vehicles (section 6.4). We finally discuss the (limited) role of changes in markups in response to the policy (section 6.5).

6.1 Exact hat algebra for the monopolistic competition and the oligopolistic cases

Suppose the reform changes the policy instrument (bonus / malus) for model j by an amount $\Delta\tau_j$ but leaves pre-tax prices unchanged. With parameters $\theta = \{\alpha, \mu_3, \mu_4\}$ and the *observed* baseline shares $\{s_j\}$, the hat-algebra yields the counterfactual share-ratio $\hat{s}_j \equiv s'_j/s_j$ as the product of three terms:

$$\hat{s}_j = \underbrace{\hat{s}_{j|s, \text{EV}}}_{\text{within segment}} \times \underbrace{\hat{s}_{s| \text{EV}}^{\circ}}_{\text{segment within EV}} \times \underbrace{\hat{s}_{\text{EV}}^{\circ}}_{\text{EV share in sales}},$$

with

$$\begin{aligned} \hat{s}_{j|s, \text{EV}} &= \frac{\exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_j]}{\sum_{k \in (s, \text{EV})} s_{k|s, \text{EV}} \exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_k]}, \\ \hat{s}_{s| \text{EV}}^{\circ} &= \frac{[\sum_{j \in s, \text{EV}} s_{j|s, \text{EV}} \exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_j]]^{\mu_3}}{\sum_{r \in S} s_{r| \text{EV}}^{\circ} [\sum_{j \in r, \text{EV}} s_{j|r, \text{EV}} \exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_j]]^{\mu_3}}, \\ \hat{s}_{\text{EV}}^{\circ} &= \frac{\sum_{r \in S} s_{r| \text{EV}}^{\circ} [\sum_{j \in r, \text{EV}} s_{j|r, \text{EV}} \exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_j]]^{\mu_3}}{\sum_{n \in \{\text{EV}, \text{ICE}\}} s_n^{\circ} \sum_{r \in S} s_{r|n}^{\circ} [\sum_{j \in r, n} s_{j|r, n} \exp[-(\alpha/\mu_{4, \text{EV}})\Delta\tau_j]]^{\mu_3}}. \end{aligned}$$

These expressions depend only on the *baseline* market shares, the estimated demand parameters, and the vector $\Delta\tau$.

¹⁹But where impacted vehicles would have not benefited from the social leasing. See discussion in section 2.1.4

Endogenous mark-ups under multiproduct Bertrand pricing To allow pre-tax prices to adjust, we close the model with the markup formula derived in Appendix C.4. Let $m^{(0)}$ denote the baseline mark-ups. Beginning with $\Delta m^{(0)} = \mathbf{0}$,

1. Compute the predicted shares $s^{(k+1)}$ using the hat expressions above with effective price change $\Delta p^{(k)} = \Delta \tau + \Delta m^{(k)}$.
2. Feed those shares into the closed-form markup equation to obtain new mark-ups $m^{(k+1)}$; the change is $\Delta m^{(k+1)} = m^{(k+1)} - m^{(0)}$.
3. Iterate $k \rightarrow k + 1$ until $\|\Delta m^{(k+1)} - \Delta m^{(k)}\| < \text{tol}$, typically a handful of iterations.

Convergence delivers a fixed point in which shares, mark-ups, and effective prices are jointly consistent with the demand system *and* Bertrand competition. In practice, the iterative solution moves the headline results only marginally relative to the fixed-price benchmark, consistent with the small incidence of the bonus on producers.

6.2 Counterfactual market shares in the absence of reform

Dropping the environmental-score (ES) filter. Table 8 shows what happens if the 2024 life-cycle screen is abolished while all other instruments stay in place. The composition of EV sales swings sharply away from China-assembled models toward European (and especially French) plants, yet total EV penetration falls by 1 percentage point. In other words, the ES filter is pivotal for reshoring value added but slows electrification at the margin.

Table 8: Counterfactual without the environmental-score reform (place of assembly)

Region	Cf with no ES reform (%)	Observed (%)	Δ (%)
Overall	17.9	16.9	-1.0
Mainland China	22.7	14.1	-8.6
Europe	49.5	56.0	6.5
Asia wo China	3.2	3.4	0.1
France	14.7	16.6	1.9
RoW	6.8	7.7	0.9

Note: Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Keeping the ES filter but canceling the 2024 CO₂ malus. Eliminating the malus while retaining the LCA threshold leaves the *within-EV* country mix essentially unchanged (Table 9); the nested structure insulates relative shares from shocks applied outside the EV nest.²⁰ What does change is

²⁰We are transparent about this feature which is somewhat at odds with findings of more sophisticated, random coefficient models. For instance, Xing et al. (2021) tend to find that EV replace proportionally more compact ICE vehicles than large ones. Our modeling choice however is consistent with the second choice data and allows for a more transparent estimation and computation of counterfactuals.

the EV adoption rate: without the tax hike on polluting ICEs, electric cars lose ground to conventional vehicles, underscoring the malus’s role as the main quantity lever.

Table 9: Counterfactual without the 2024 malus increase (assembly mix among EVs)

Region	Cf with no MALUS reform (%)	Observed (%)	Δ (%)
Overall	16.0	16.9	0.9
Mainland China	14.1	14.1	0.0
Europe	56.0	56.0	0.0
Asia wo China	3.4	3.4	0.0
France	16.6	16.6	0.0
RoW	7.7	7.7	0.0

Note: Counterfactual evaluated under the no 2024 malus increase (applying 2023 schedule) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Battery-cell sourcing. Relaxing the ES constraint has a large impact upstream in the battery supply chain (Table 10). The model predicts about a 5 pp decline in Chinese market share with a rise in EU-made battery cells of about 4 pp.

The model predicts that the reform increased EU-made battery purchases by about 10,168 units. There are two countervailing forces. Holding the EV market share constant, the rise in the EU share within EVs would have raised EU-made battery demand by about 12 300 units, while the contraction in overall EV demand subtracts about 2 100 units.

Table 10: Counterfactual without the environmental-score reform (battery-cell origin)

Region	Cf with no ES reform (%)	Observed (%)	Δ (%)
Overall	17.9	16.9	-1.0
Mainland China	58.7	54.7	-4.1
Europe	31.6	35.9	4.3
Asia wo China	6.0	6.6	0.6
RoW	0.1	0.1	0.0
Unknown	0.6	0.7	0.1

Note: Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

The ES screen and the CO₂ malus act on different margins. The former reallocates demand toward lower-carbon—and more domestic—supply chains but suppresses EV uptake, whereas the latter accelerates electrification without altering the geography of value added. This result is a direct consequence of the nested logit assumption so it needs to be interpreted with caution. Overall, it suggests, however, that the two policies are complementary.

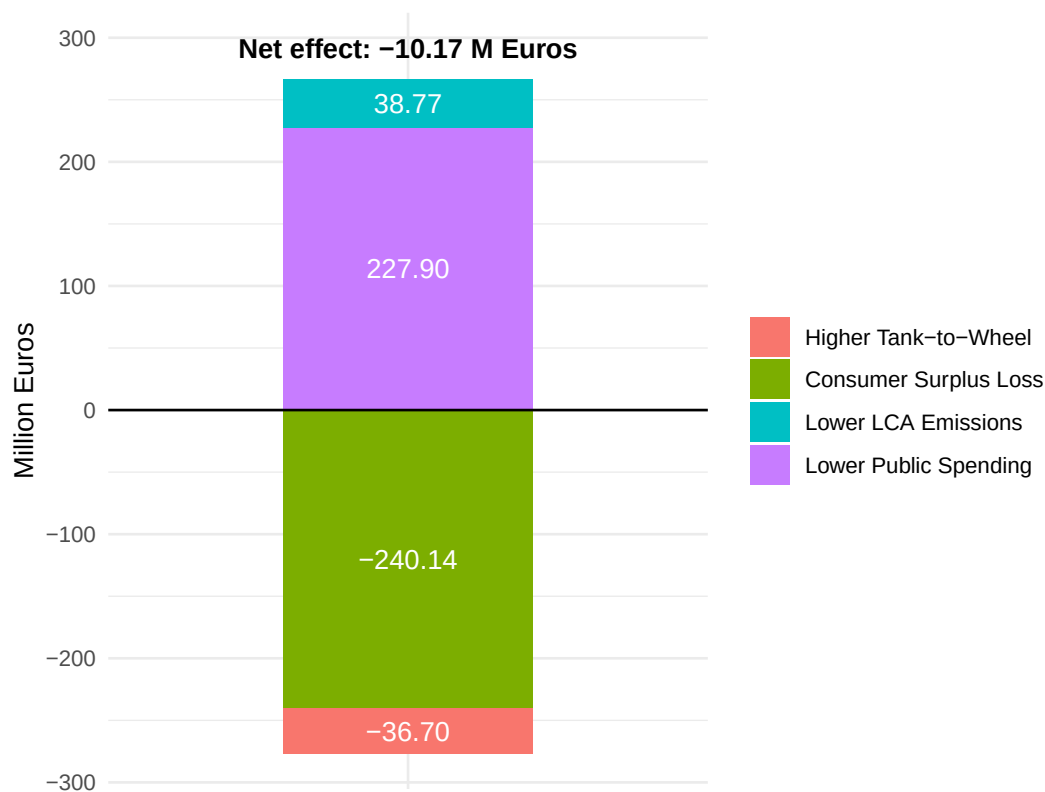
6.3 Cost benefit analysis

We now turn to the welfare implications of the reform. Our cost-benefit analysis considers two measurable costs and two measurable benefits, all expressed in monetary terms. On the cost side: (i) the reform tightens eligibility, raising average consumer prices and reducing consumer surplus; (ii) by slowing electrification, it increases tank-to-wheel CO₂ emissions from the fleet. On the benefit side: (i) tighter eligibility reduces public spending on the bonus; (ii) the life-cycle screen reallocates demand toward models with lower manufacturing-phase emissions, reducing upstream CO₂. We value both types of environmental externalities at a social cost of carbon of 200 EUR/tCO₂ and assume a vehicle lifetime of 180,000 km (see Appendix C.6 for a full description of the parameters). In this first accounting, we hold pre-tax prices fixed and set aside producer surplus; we return to both in the AVPF analysis below and in section 6.5.

Beyond these four channels, the reform may generate industrial pay-offs that are harder to monetize: scale economies in European battery production, learning in local assembly, and option value for future competitiveness. We do not attempt to quantify these gains directly. Instead, by computing the net cost per additional European-sourced battery, our analysis provides an implicit price tag on the reshoring objective—allowing the reader to assess whether the industrial benefits are plausibly large enough to justify the welfare cost.

Quantification. Figure 12 assembles all quantified items. Consumer surplus falls by about €240 million while tank-to-wheel emissions increase by an amount valued at €37 million. On the benefit side, public outlays on the bonus fall by an estimated €228 million—largely the mirror image of the consumer surplus loss—and the manufacturing-phase externality gain is worth about €39 million. The net balance of $[240 + 37 - 228] = \text{EUR } 49$ million implies a cost of roughly €4,800 per additional EV whose battery is sourced in Europe. When the manufacturing-phase externality benefit is included, that figure falls to about €1,000.

Figure 12: Decomposition of the net welfare effect of the 2024 life-cycle-based eligibility reform.



Note: Positive bars represent social benefits, negative bars social costs. Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Average Value of Public Funds. The environmental-score reform restricts subsidies: it saves public money by excluding high-LCA models from the bonus. Following Hahn et al. (2026), we evaluate its fiscal efficiency through the Average Value of Public Funds (AVPF), defined as the ratio of aggregate willingness to pay to net fiscal savings ($-\Delta G$). Because the reform is non-marginal—it removes a subsidy from about one-third of the EV market overnight—we compute an *average* rather than marginal value of public funds. Because the reform economizes public spending, an AVPF below one indicates that the willingness to pay to avoid the spending cut is less than the fiscal savings—in other words, the reform is an efficient way to raise public funds. Table 11 presents the results under two perspectives (global and local) and with or without fiscal externalities.

From a global standpoint—counting consumer surplus, environmental externalities, and total producer surplus against fiscal savings—the AVPF is 1.07 without fiscal externalities: the social cost of the reform slightly exceeds the fiscal savings. From the perspective of French citizens, the local AVPF is 0.82—below one—reflecting the fact that a substantial share of the producer surplus loss falls on foreign firms and is therefore excluded from the local welfare calculus.

Accounting for fiscal externalities changes the picture substantially. By slowing electrification, the reform increases ICE driving and thus generates additional fuel-tax (TICPE) revenue over the vehicles’ lifetime. This channel is first-order for the welfare assessment: the global AVPF falls from 1.07 to 0.75, and the local AVPF from 0.82 to 0.58, once fuel-tax revenue and corporate tax on French-brand profits are included. As discussed in the introduction, the reform effectively converts largely untaxed carbon (manufacturing-phase emissions) into heavily taxed carbon (driving emissions, taxed via TICPE at approximately €250/tCO₂)—a pre-existing wedge that mechanically inflates the fiscal return of the reform. Appendix Table C.5.1 presents the counterpart under Bertrand pricing, with very similar results.

Table 11: AVPF of the actual reform (fixed prices).

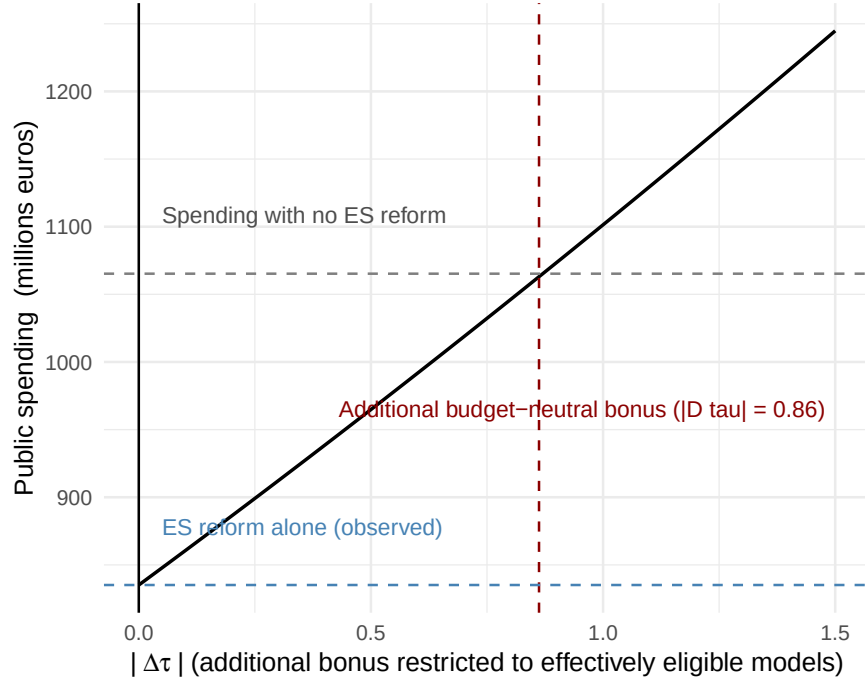
	W/o fiscal ext.	W/ fiscal ext.
Numerator		
Global: $\Delta CV + \Delta EX_{TTW} + \Delta EX_{LCA} + \Delta PS$	-275	-275
Local: $\Delta CV + (1 - t) \theta_{FR} \Delta PS_W$	-219	-219
Denominator ($-\Delta G$)		
Bonus savings	-270	-270
Fuel tax (TICPE)	—	-106
Corporate tax on FR PS	—	-12
Global AVPF	1.02	0.71
Local AVPF	0.81	0.56

Note: Fixed-price counterfactual. Global numerator: $\Delta CV + \Delta EX_{TTW} + \Delta EX_{LCA} + \Delta PS$. Local numerator: $\Delta CV + (1 - t) \theta_{FR} \Delta PS_W$ where $\theta_{FR} = 20\%$ and $t = 25\%$. Denominator: net fiscal savings (bonus spending change plus, when applicable, fuel tax and corporate tax).

6.4 Simple budget neutral counterfactual policy

Policy design The actual 2024 environmental-score (ES) reform cut public spending by excluding high manufacturing-phase emission models from the bonus. We consider a *budget-neutral* variant that keeps the ES eligibility filter in place but recycles the saved funds into a uniform top-up for the *remaining* eligible EVs. The revenue-neutral condition implies an average increase of $|\Delta\tau| \approx e900$ per qualifying vehicle (Fig. 13).

Figure 13: Public spending before and after re-allocating the savings from the environmental-score (ES) eligibility filter.



Note: Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Welfare decomposition. The larger bonus shifts an extra +14,320 cars toward Europe-sourced batteries while holding the aggregate EV share roughly constant. Consumer surplus declines by €61 million, so the implicit cost is $CV/\Delta\text{battery}_{EU} \approx \text{€}4,200$ per additional “reshored” battery—about one-third of the battery’s production cost (Fig. 14). The cost per reshored battery depends, however, on how producer surplus and fiscal externalities are accounted for. Table 12 summarizes the results.

Under the assumption that approximately 20% of the profits of winning firms accrue to French citizens (through equity ownership),²¹ and accounting for the corporate tax on profits of French-

²¹The 20% ownership share is a weighted average across winning firms. Stellantis (66% of winner PS) is approximately 21% French-owned: Peugeot Invest 7.1%, BPI France 6.1%, Amundi 2.7%, plus estimated other French institutions and retail (~5%) (source: Stellantis 2024 Annual Report; MarketScreener). Renault (13% of winner PS) is approximately 47% French-owned: French state 15%, French institutional investors 11.4%, employees 5.7%, estimated French retail ~15%

brand subsidiaries,²² the net cost to French citizens falls from €4,200 to approximately €1,600 per reshored battery. The reform also generates fiscal externalities beyond the (zero) change in bonus spending: fewer EVs imply more ICE driving and thus additional fuel tax (TICPE) revenue of approximately €11 million over the vehicles' lifetime.²³ Including this fiscal channel and corporate tax on French-brand profits, the cost per reshored battery falls to approximately €800. Overall, the same reform appears costly from a pure consumer welfare standpoint (€4,200/battery), but near welfare-neutral when domestic producer gains and fiscal externalities are included (€800/battery).

Table 12: Budget-neutral reform: welfare decomposition (fixed prices).

Component	M EUR	Local (FR)	€/battery
Consumer surplus (ΔCV)	-61	-61	4,500
Environmental TTW (SCC=200)	-24		
Environmental LCA (SCC=200)	+51		
$(1 - t) \theta_{FR} \Delta PS_W$		+30	
Corp. tax on FR-brand PS (55%)		+16	
Fuel tax (TICPE)		+26	
Total ΔPS (all firms)	+34		
Net welfare (global)	1		100
Local (excl. env.)		-14	1,000
Local (incl. env.)		13	1,000
Local + fiscal ext.		11	net gain

Note: Fixed-price counterfactual (monopolistic competition). $\theta_{FR} = 20\%$ (weighted average of French equity ownership across winning firms: Stellantis $\approx 21\%$, Renault $\approx 47\%$, others $\approx 0\%$). Corporate tax at $t = 25\%$ applied to 55% of French-brand PS (share taxable in France given assembly locations). Fuel tax computed using TICPE $\approx \text{€}0.036/\text{km}$, lifetime = 180,000 km. Appendix Table C.5.2 presents the counterpart under Bertrand pricing.

Environmental balance. Because the tightened eligibility still excludes high-LCA imports, the life-cycle gain per vehicle is preserved, while the higher subsidy prevents the drop in fleet electrification. As a result, the budget-neutral variant dominates the original reform on both decarbonization and welfare grounds, albeit at a non-negligible consumer cost.

An important asymmetry reinforces this conclusion. The budget-neutral reform reduces manufacturing-

(source: Renault URD 2023; Zonebourse). Other winning firms (VW Group, BMW, Hyundai, Toyota) have negligible French ownership. Weighted: $(0.66 \times 21 + 0.13 \times 47) / 100 \approx 20\%$.

²²We apply the 25% French corporate tax rate to 55% of the change in French-brand profits (Renault + Stellantis's Peugeot, Citroën, DS, Dacia, Alpine brands). The 55% reflects the share of French-brand profits taxable in France: 31% of French-brand revenue comes from vehicles assembled in France (where 100% of the markup is French-taxed), and the remaining 69% generates only distribution margins in France (approximately 35% of the total markup). Stellantis N.V. is Dutch-resident, but its French-brand subsidiaries are French tax entities. Stellantis's non-French brands (Fiat, Opel, Jeep) are excluded.

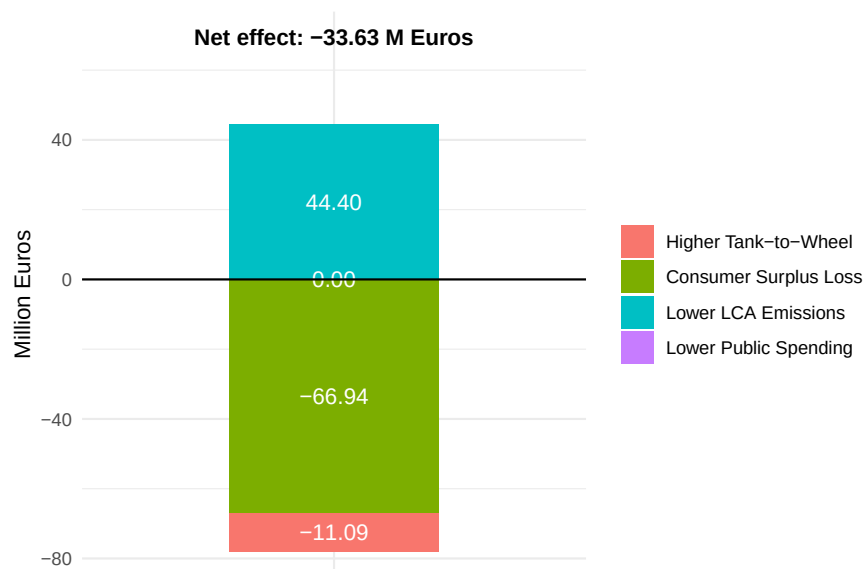
²³TICPE implies an implicit carbon price of approximately €250/tCO₂, above the SCC of 200 EUR/tCO₂ used to value the environmental externality.

phase (life-cycle) emissions by €44 million while increasing tank-to-wheel emissions by only €11 million—a net environmental gain. Because these two types of emissions face very different levels of taxation in France (see the discussion in Section 6.3), the net environmental benefit of the reform is plausibly larger than what a uniform social cost of carbon would imply.

Abatement cost. We can express the welfare cost of the budget-neutral reform per ton of net CO₂ abated. The reform avoids approximately 255,000 tons of manufacturing-phase emissions while generating roughly 95,000 additional tons from driving—a net reduction of about 160,000 tCO₂. From a pure consumer surplus standpoint, the implied abatement cost is approximately €380/tCO₂, above the social cost of carbon of 200 EUR/tCO₂ used in our calculations. From a local welfare perspective (accounting for domestic producer surplus and corporate tax but excluding fiscal externalities), the abatement cost falls to about €140/tCO₂—below the SCC. Including the fuel-tax fiscal externality, the abatement cost is close to zero: the TICPE revenue from the additional ICE driving effectively finances the manufacturing-phase emission reductions.

For comparison, Gillingham and Stock (2018) report abatement costs for EV subsidies in the range of \$100–\$300/tCO₂, and Holland et al. (2016) find wide variation across US states (from negative values to over \$2,800/tCO₂). Our consumer-only estimate of €380/tCO₂ is in the upper range of these benchmarks. Both studies, however, focus on the abatement of tank-to-wheel or grid-related emissions from electric driving—a margin that is likely very small in France given its largely decarbonized electricity grid. In our setting, the relevant margin is the reduction of manufacturing-phase emissions, which neither study considers.

Figure 14: Decomposition of the net welfare effect for the budget-neutral variant of the ES reform.



Note: Positive bars represent social benefits, negative bars social costs. Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *fixed mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

6.5 Producer-surplus effects and variable markups

Up to this point the counter-factual experiments have held pre-tax prices constant, consistent with monopolistic competition with CRS. We now resolve every scenario under multiproduct Bertrand competition: for each firm f , $(\Theta_f \odot \Delta) (p_f - c_f) = -s_f$, with Θ_f the ownership matrix and Δ the Jacobian of market shares (see App. C.4). Mark-ups are re-evaluated after the policy shock and shares and prices are iterated to convergence.

As can be seen in Table 13, allowing mark-ups to adjust has a negligible impact on the reallocation away from China and on the aggregate fall in EV demand when compared with the constant-mark-up benchmark (see Table 8). Table 14 shows that the shift of battery demand from China to Europe is virtually identical to the constant-mark-up case, confirming that changes in markups do not materially affect sourcing patterns.

Table 13: Counterfactual *without* the environmental-score reform: EV market shares by place of assembly *with* endogenous mark-ups.

Region	Cf with no ES reform (%)	Observed (%)	Δ (%)
Overall	18.0	16.9	-1.1
Mainland China	22.3	14.1	-8.2
Europe	49.8	56.0	6.2
Asia wo China	3.2	3.4	0.2
France	14.8	16.6	1.9
RoW	6.9	7.7	0.8

Note: Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *variable mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Table 14: Counterfactual *without* the environmental-score reform: EV market shares by battery-cell origin *with* endogenous mark-ups.

Region	Cf with no ES reform (%)	Observed (%)	Δ (%)
Overall	18.0	16.9	-1.1
Mainland China	58.7	54.7	-4.0
Europe	31.7	35.9	4.1
Asia wo China	5.9	6.6	0.7
RoW	0.1	0.1	0.0
Unknown	0.6	0.7	0.1

Note: Counterfactual evaluated under the *lower-bound shock* scenario (see section 2.1.4) with *variable mark-ups* (see section 6.5 for more on varying markups). Appendix section C.6 provides details on all parameters used in the simulation.

Quantitatively, mark-ups change by less than €100 on average. Under fixed prices, overall producer surplus falls by about €26 million.²⁴ The headline redistribution is internal: firms with

²⁴Under Bertrand pricing, the sign reverses: total producer surplus rises slightly as winning firms adjust markups upward. The redistribution across firms is qualitatively identical.

positive change in surplus experience €164 million in gains while losing firms total €-139 million in losses.

Because mark-ups hardly move, most of the statutory subsidy is captured by buyers rather than sellers. With prices determined by $p = c + m + \tau$, total pass-through is $dp/d\tau = 1 + dm/d\tau$. Empirically we find $dm/d\tau \approx 0$, implying full incidence on consumers—in line with reduced-form evidence on listed prices—and confirming that the reform’s welfare costs are borne almost exclusively on the demand side.

7 Conclusion

We assess France’s 2024 “score environnemental”, i.e. life-cycle bonus–malus reform. Reduced-form estimates show registration volumes respond strongly to changes in incentives. Beyond substitution among EVs, the reform flattened the upward EV trend as our structural model attributes a 0.4-percentage-point drop in 2024 EV share to the environmental score. Our counterfactual analysis suggests that tank-to-wheel CO₂ costs exceed life-cycle gains in the baseline counterfactual. We show that a simple budget-neutral redesign would be carbon-positive while still reshoring battery demand substantially. The cost per reshored battery—approximately €4,200 from consumer surplus alone, falling to roughly €1,600 when domestic producer gains are accounted for—sets an implicit price tag on the reshoring objective: industrial benefits would need to clear this bar to justify the welfare cost. The welfare assessment is further shaped by a pre-existing asymmetry in carbon pricing: the reform curtails largely untaxed manufacturing-phase emissions while modestly expanding driving emissions that are already taxed well above the social cost of carbon through fuel excise. The distributional consequences of the policy depend greatly on where the buyers of EVs that became excluded by the reform sit on the income distribution. In the absence of appropriate micro-data, this is a difficult question to tackle but remains an important avenue for further research.

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Online Appendix

Table of contents

A Data appendix	OA-1
A.1 Determining Country of Battery Origin	OA-1
A.2 Other elements	OA-1
B Empirical appendix	OA-1
B.1 Synthetic control analysis	OA-1
B.2 Bonus: Robustness and Further Event Studies	OA-5
B.3 Malus: Robustness and Further Event Studies	OA-8
B.4 French second-choice survey: substitution heatmaps and moments	OA-10
C Model appendix	OA-15
C.1 Correlation structure of the nested logit	OA-15
C.2 Nesting order: fuel-type first vs. segment first	OA-15
C.3 Inference for the nest scale parameters	OA-17
C.4 Markups in the nested logit model	OA-19
C.5 Welfare tables under Bertrand pricing	OA-20
C.6 Parameters used for the cost benefit analysis	OA-22

A Data appendix

A.1 Determining Country of Battery Origin

In order to determine the origin of the battery of a model identified at Brand x Model level j , a two step procedure is implemented. A first data set allows us to identify the number of units of j sold in France by country of assembly. Let S_{jz}^A be the number of units of Brand x Model j sold in France and assembled in z . The share of sales of j in France that were assembled in country z then is:

$$s_{jz}^A = \frac{S_{jz}^A}{\sum_{z=1}^Z S_{jz}^A}$$

A second data set allows us to identify the world wide sales of model j assembled in country z by the location of battery production k . Let S_{jzk}^B be the number of units of Brand x Model j sold worldwide, assembled in country z with a battery from country k . The share of sales of j with battery from k in world wide sales of j assembled in country z is then:

$$s_{jzk}^B = \frac{S_{jzk}^B}{\sum_{k=1}^K S_{jzk}^B}$$

Since we only observe the world wide sales of units of j with batteries from different countries k , we approximate the share of units of j sold in France with batteries from k as follows:

$$s_{jk} = \sum_{z=1}^Z s_{jz}^A * s_{jzk}^B$$

A.2 Other elements

B Empirical appendix

B.1 Synthetic control analysis

In order to better understand changes in the electric vehicle market in Europe we employ the synthetic control method (SCM) as developed in Abadie and Gardeazabal (2003), Abadie et al. (2010) and Abadie (2021). We analyse both the change in EV subsidies in France and the abolition of subsidies in Germany.

Data. For this analysis, we use monthly data on electric vehicle market shares in 27 European countries, provided by the European Alternative Fuels Observatory. This data is available from January 2020 until August 2024 (more requested).

Figure A.2.1: Two models disqualified by the 2024 ES (battery and assembly in China).



Note: The Dacia Spring (left) and MG4 (right) both lost eligibility for the French EV bonus in January 2024 due to high manufacturing-phase emissions. Both are assembled in China with Chinese-sourced battery cells. The Dacia Spring was the #2 best-selling EV in France in 2023 (32k units); the MG4 was #6 (22k units).

Basic setup. The goal of the SCM is to generate basic counterfactuals for how the EV market would have developed absent any reform. In order to achieve this, the pre-reform trends in countries without any reforms (donor pool) are weighted in such a way as to most closely track the pre-reform trend of the country that experiences the reform. The weights are optimized for 2023 in order to align the synthetic trend with the observed market shares. Comparing the weighted trend post-reform with the observed trend then allows one to better understand whether a reform differentially impacted the affected country. The donor pool for France does not include Germany and vice versa. Furthermore, the following countries were removed from the donor pool since they themselves experience major policy changes around January 2024: Belgium, Italy and Ireland. The weights resulting from the SCM are presented in Table B.1.1. Figure B.1.1 below plots the observed and counterfactual trends for France and Germany. While initially France seems to fare better than its synthetic counterfactual, the trends quickly align again making it difficult to identify an effect. The same analysis for Germany on the other hand shows a clear difference between the observed and synthetic trend, with the observed market share underperforming its counterfactual.

Table B.1.1: Synthetic control weights

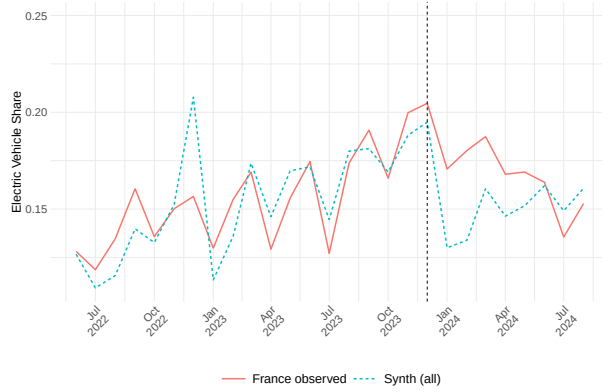
Donor	France	Germany
Netherlands	0.20	0.25
Spain	0.62	0.57
Sweden	0.18	0.18

In line with Grassi (2024), we employ two robustness exercises presented in the next section.

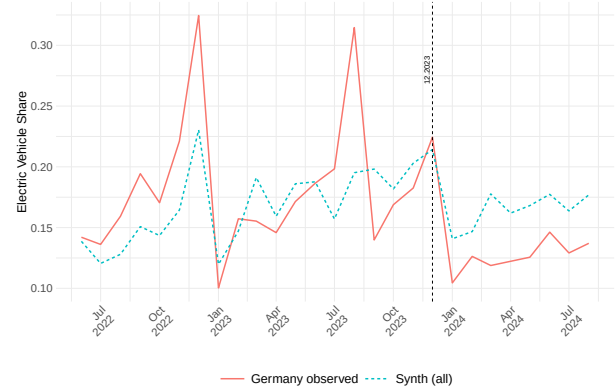
Leave-one-out robustness test: here we remove each non-zero weight donor one by one from the donor pool and reestimate the weights. No matter which of the positively weighted donors is removed, the overall picture remains largely unchanged.

In-country-placebo robustness test: here we plot the difference between the observed and counterfactual trends for all countries in the donor pool. While the gap between the observed French and synthetic French trend crosses into the interquartile range of other European countries, the German gap stays below it.

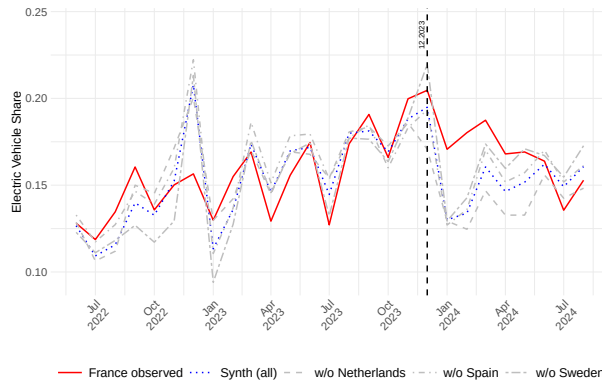
Figure B.1.1: Synthetic control results and robustness checks for France and Germany.



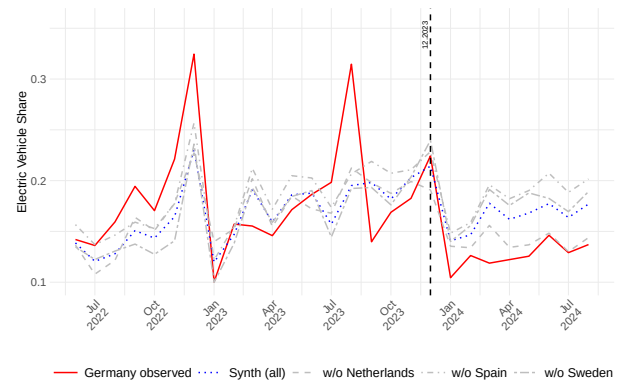
Synthetic control: France



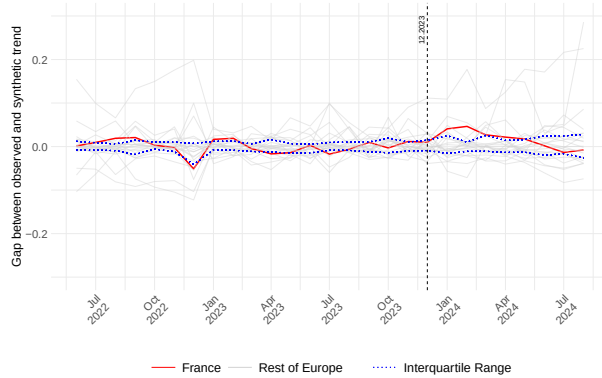
Synthetic control: Germany



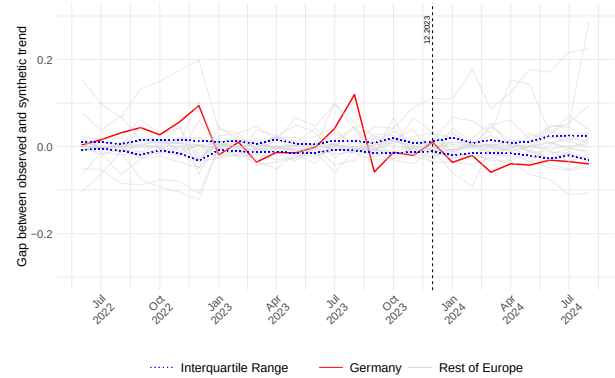
Leave-one-out: France



Leave-one-out: Germany



Placebo test: France



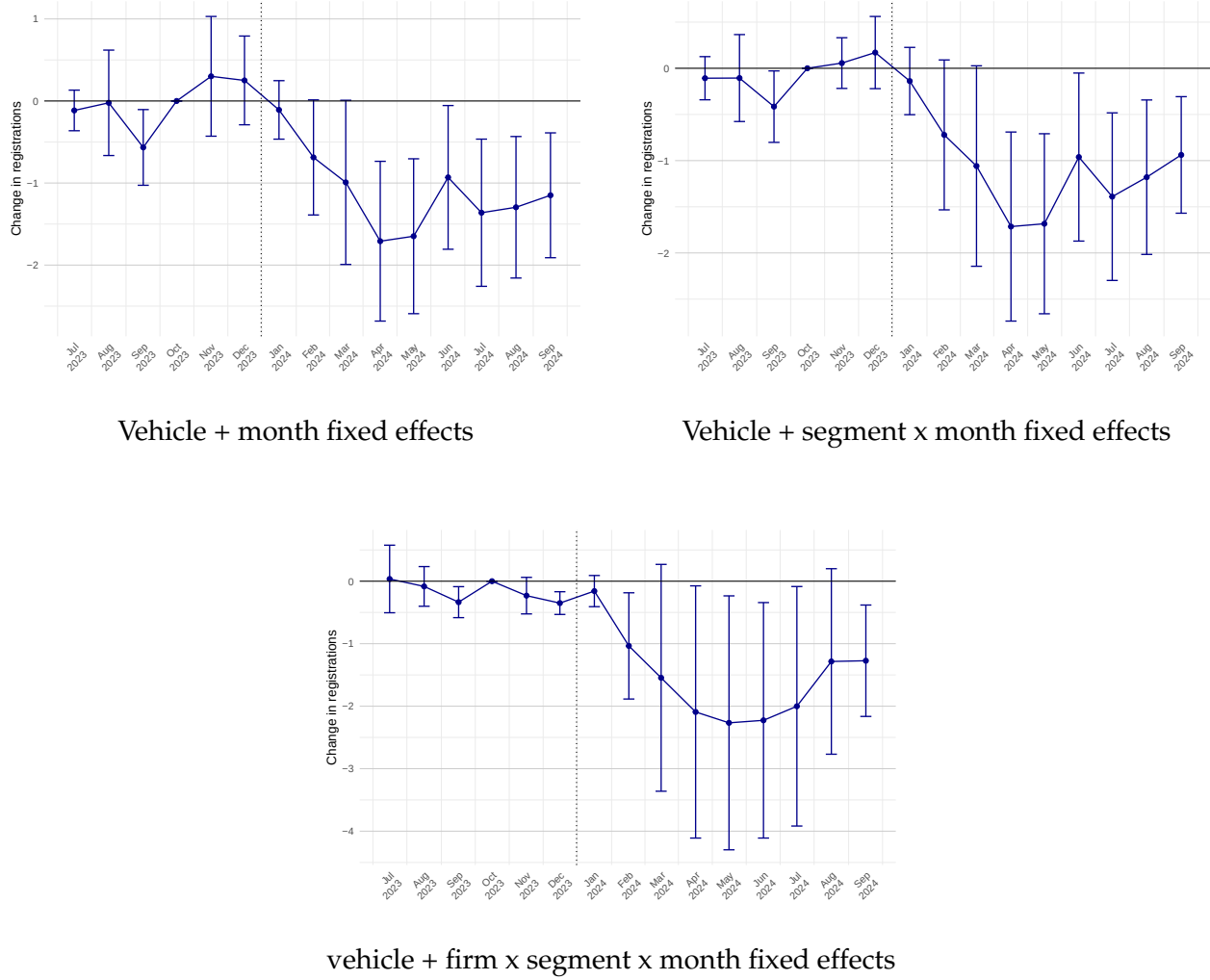
Placebo test: Germany

Note: Top row: synthetic control estimates of the effect of the January 2024 policy changes on EV market shares in France (left) and Germany (right). Bottom row: placebo tests using the same donor pool. The vertical dashed line marks the policy change date. The synthetic control is constructed from a donor pool of EU countries using pre-treatment EV market shares as predictors. See Appendix B.1 for details on the methodology and donor weights.

B.2 Bonus: Robustness and Further Event Studies

B.2.1 Brand x Model x PF aggregation

Figure B.2.1: Event study of loss of eligibility among all EVs



Notes: This figure presents results of a dynamic difference-in-differences regression of number of new vehicle registrations on loss of eligibility according to equation 3. Vertical bars indicate a 95% confidence interval. Monthly observations are at the Brand x Model x PF level. The main specification can be found in figure 8.

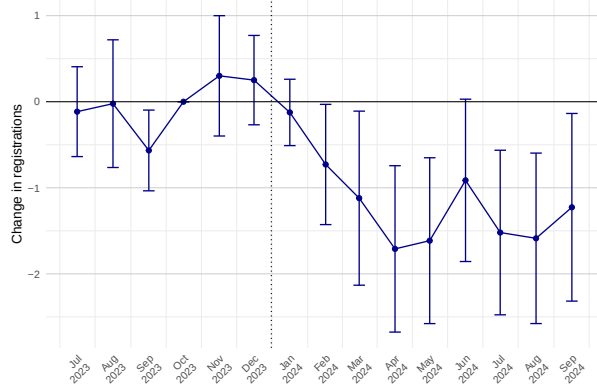
Table B.2.1: Static regression of number of new vehicle registrations on loss of eligibility in 1000 of euros (cnit level)

Dependent Variable:	Market Share		
Model:	(1)	(2)	(3)
<i>Variables</i>			
Treatment (TP)	-0.126*** (0.045)	-0.140*** (0.042)	-0.246*** (0.051)
<i>Fixed-effects</i>			
Carline	Yes	Yes	Yes
Month	Yes		
Month $\times$ Segment		Yes	
Month $\times$ Segment $\times$ Firm			Yes
<i>Fit statistics</i>			
Observations	8,760	8,760	8,555
Pseudo R ²	0.265	0.267	0.282
Implied own-price elasticity	-3.67	-4.09	-7.19

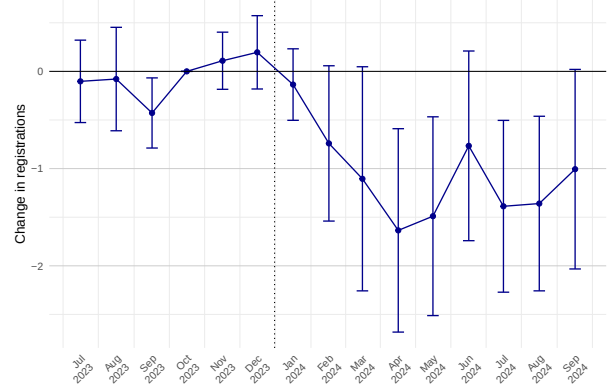
Notes: This table presents results of a static difference-in-differences regression of number of new vehicle registrations on loss of eligibility according to equation 4. Coefficients capture the effect of a 1000 euro price increase. Observations are grouped at the cnit level. The same regression with observations grouped at the Brand $\times$ Model $\times$ Power level can be found in table 3. Clustered (carline) standard-errors in parentheses. Signif. Codes: ***, 0.01, **, 0.05, *, 0.1.

B.2.2 No aggregation (cnit level analysis)

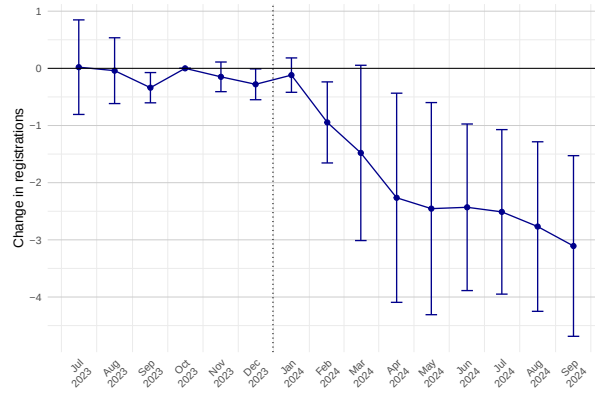
Figure B.2.2: Event study of loss of eligibility among all EVs at the CNIT level



Vehicle + month fixed effects



Vehicle + segment x month fixed effects



Vehicle + firm x segment x month fixed effects

Notes: This figure presents results of a dynamic difference-in-differences regression of number of new vehicle registrations on loss of eligibility according to equation (3). Vertical bars indicate 95% confidence intervals. **Monthly observations are at the CNIT level.** The main specification can be found in Figure 8.

B.3 Malus: Robustness and Further Event Studies

B.3.1 Malus: Further Static Regressions

Table B.3.1: Controlling for model-implied changes in markups in the main DiD analyses

Model:	Baseline ICE (1)	Iterated ICE (2)	Baseline EV (3)	Iterated EV (4)
<i>Variables</i>				
Malus (per €1k)	-0.1776*** (0.0442)	-0.1775*** (0.0447)		
Bonus Loss (per €1k)			-0.1563*** (0.0418)	-0.1564*** (0.0418)
<i>Fixed-effects</i>				
Carline	Yes	Yes	Yes	Yes
Month $\times$ Segment $\times$ Motorisation	Yes	Yes		
Month $\times$ Segment			Yes	Yes
<i>Fit statistics</i>				
Observations	23,717	23,717	2,124	2,124
Pseudo R ²	0.20851	0.23545	0.23405	0.28449

Notes: This table presents results of a static difference-in-differences regression of market share on continuous malus increase according to equation 4 in column (1) and (2). Column (1) is equivalent to column (2) in table 5, except that it includes all non-EV cars (hybrid and plug-in vehicles in addition to petrol and diesel), which accounts for the difference in coefficients. Column (3) is equivalent to column (2) of Table 3. Column (2) controls for estimated markups among all non-EVs. Column (3) and (4) are the EV counterpart to columns (1) and (2). The time period considered covers six pre- and six post-reform months. Observations are grouped at the brand $\times$ model $\times$ Co2 g/km level for ICEs and brand $\times$ model $\times$ power level for EVs, where power is measure by puissance fiscale.

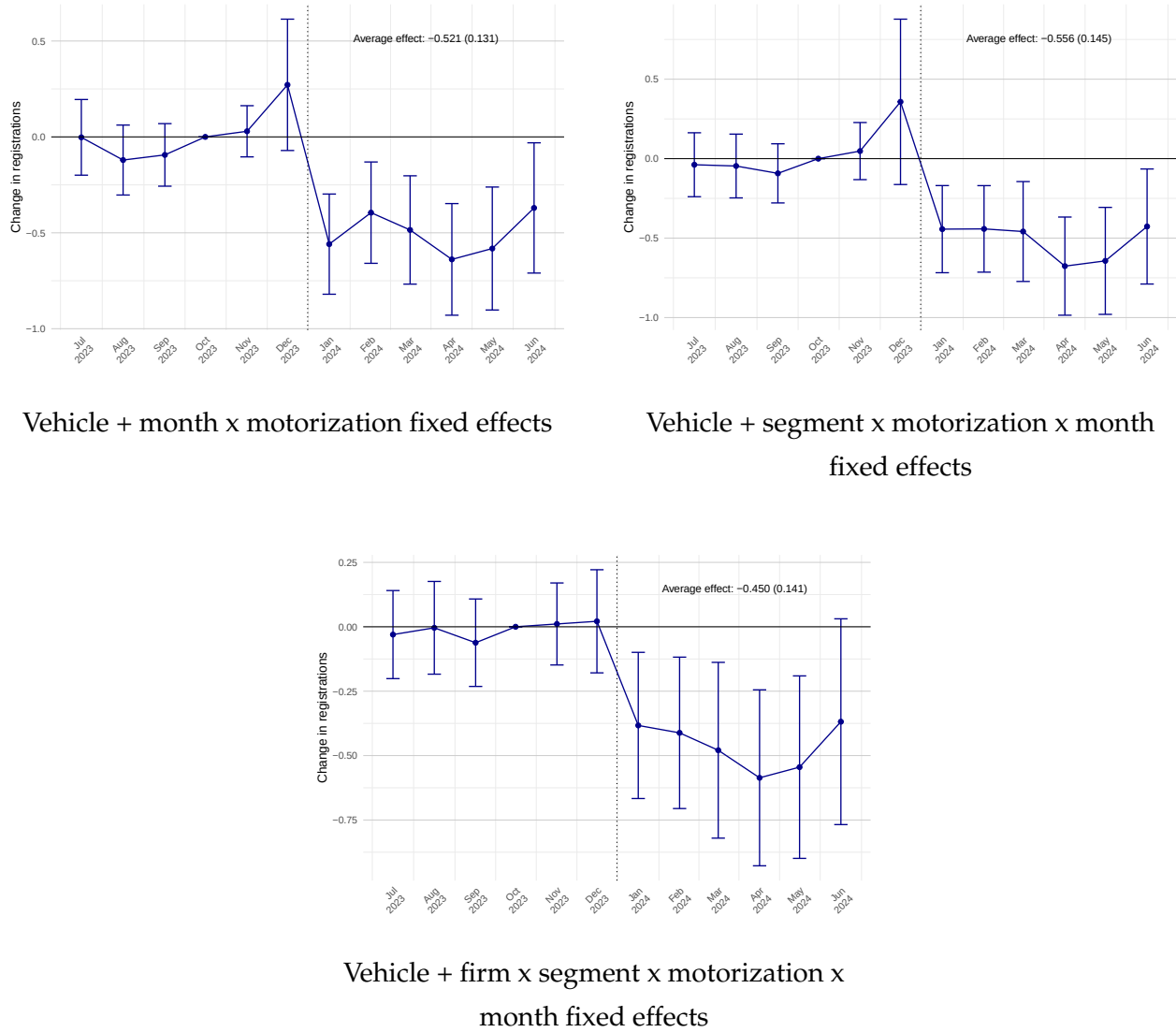
Table B.3.2: Static regression among ICE of market share on binary indicator for malus increase bigger than 500 euro

Dependent Variable: Model:	Market Share		
	(1)	(2)	(3)
<i>Variables</i>			
$\mathbb{1}\{\Delta\tau_j > 500\} \times Post$	-0.521*** (0.131)	-0.556*** (0.145)	-0.450*** (0.141)
<i>Fixed-effects</i>			
Month $\times$ Motorisation	Yes		
Carline	Yes	Yes	Yes
Month $\times$ Segment $\times$ Motorisation		Yes	
Month $\times$ Segment $\times$ Firm $\times$ Motorisation			Yes
<i>Fit statistics</i>			
Observations	23,927	23,717	23,058
Pseudo R ²	0.23053	0.23012	0.23010

Note: This table presents results of a static difference-in-differences regression of market share on a binary indicator for malus increase larger than 500 euros according to equation 4. The time period considered covers six pre- and six post-reform months. Observations are grouped at the brand $\times$ model $\times$ Co2 g/km level. Clustered (carline) standard-errors in parentheses. Signif. Codes: ***: 0.01, **: 0.05, *: 0.1.

B.3.2 Malus: Event Studies

Figure B.3.1: Event study of increase in malus among ICE (binary treatment: malus increase > 500 euros)



Note: This figure presents results of a dynamic difference-in-differences regression of within-segment market share on a binary indicator for malus increase larger than 500 euros according to equation (3). Vertical bars indicate 95% confidence intervals. Observations are grouped at the Brand $\times$ Model $\times$ CO₂ g/km level. Each panel uses a different set of fixed effects.

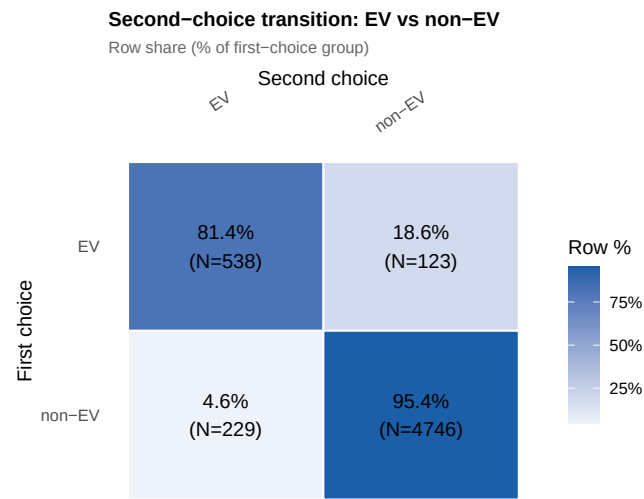
B.4 French second-choice survey: substitution heatmaps and moments

We conducted a survey in May 2026 ($N = 5,645$ complete responses, representative of recent French car buyers) in order to collect stated preferences on second choice preferences. Each respondent stated their current or most recent car (first choice) and indicated which vehicle they would have

purchased instead had their preferred model been unavailable (second choice). Motorisation type (electric, plug-in hybrid, hybrid, petrol, diesel) was recorded directly by the survey questionnaire.

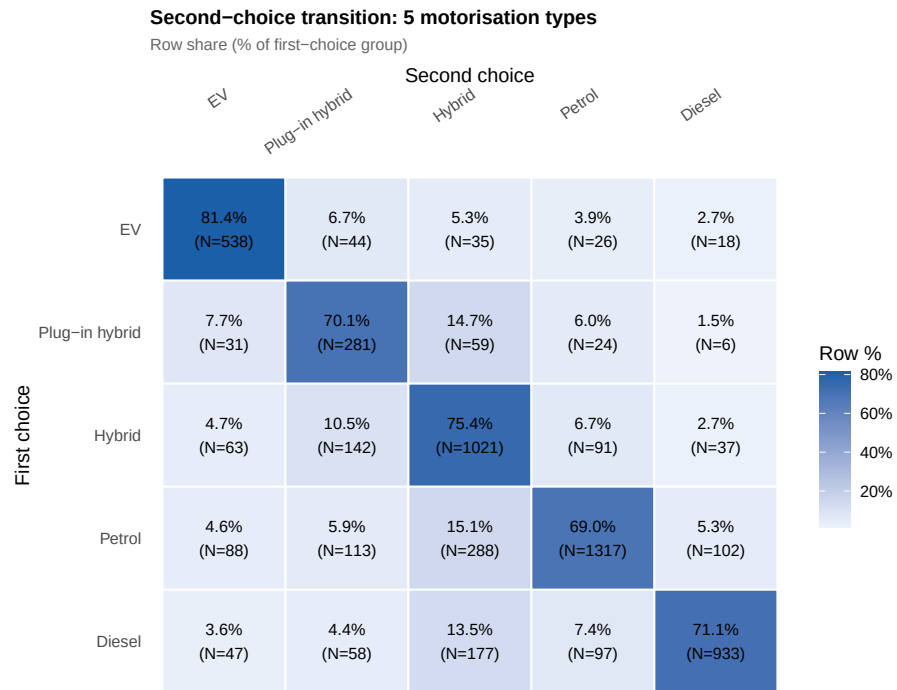
Substitution heatmaps. Figures B.4.1–B.4.4 visualise the substitution matrix as row-normalised shares. Each row sums to 100%: the diagonal reports the *retention rate* within a motorisation type or segment. Figure B.4.1 confirms that EV buyers are highly likely to stay within the EV nest (81.4%), while non-EV buyers almost never switch to EVs as a second choice (only 4.9%). Figures B.4.2 disaggregates this by all five motorisation categories. Figures B.4.3–B.4.4 show segment-level substitution patterns for all respondents and for EV first-choice owners respectively; they reveal moderate within-segment loyalty (roughly 30–45% for most segment/motorisation combinations among EV buyers).

Figure B.4.1: Second-choice transitions: EV vs non-EV (row %)



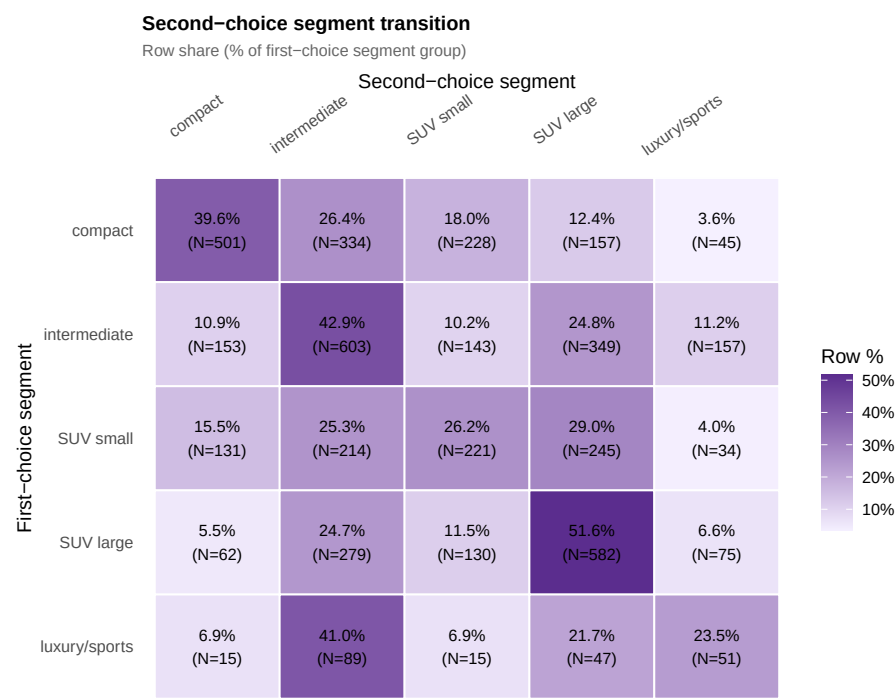
Note: Row-normalised transition matrix from first choice (rows) to second choice (columns). Colours scale with row share. $N = 5,645$ complete responses, French stated-preference survey, May 2026.

Figure B.4.2: Second-choice transitions: five motorisation types (row %)



Note: Row-normalised transition matrix across five motorisation types. Each row sums to 100%. French stated-preference survey, May 2026.

Figure B.4.3: Second-choice segment transitions — all respondents (row %)

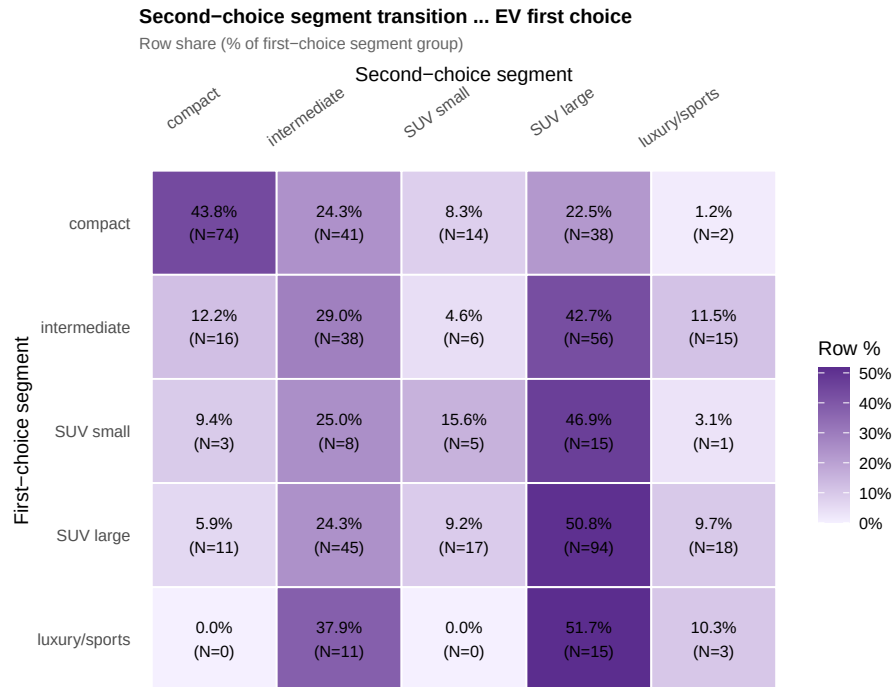


Note: Row-normalised segment-level transition matrix for all respondents with both first- and second-choice segment matched (93.6% of sample). Segments: compact, intermediate, small SUV, large SUV, luxury /sports.

Table B.4.1: Second-choice moments from the French survey (May 2026, $N = 5,645$). Rows marked [untargeted] are used for out-of-sample validation only.

Moment	Share	N
X_{EV}^o : $P(EV \rightarrow EV)$	0.814	661
$X_{s,EV}^o$: $P(EV \rightarrow EV, \text{ same segment})$	0.332	614
X_s^o : $P(\text{same segment, all buyers})$	0.402	4871
X_{ICE}^o : $P(\text{non-EV} \rightarrow \text{non-EV})$ [untargeted]	0.954	4975
$X_{s,ICE}^o$: $P(\text{non-EV} \rightarrow \text{non-EV, same segment})$ [untargeted]	0.386	4636

Figure B.4.4: Second-choice segment transitions — EV first-choice owners (row %)



Note: Row-normalised segment-level transition matrix restricted to EV first-choice owners ($N = 447$ with segment matched). Reveals within-segment retention conditional on the EV nest.

Target moments for the structural model. Table B.4.1 reports the three aggregate moments that enter the structural estimation as identification targets for μ_3 and $\mu_{4,EV}$. The French EV retention rate (0.814) is substantially higher than the US estimate from Allcott et al. (2024) (0.52), consistent with the higher EV penetration and stronger within-nest loyalty we observe in France by 2025–2026. The EV same-segment retention rate (0.332) is close to the US estimate (0.32).

C Model appendix

C.1 Correlation structure of the nested logit

The nested logit arises from a random utility model in which the idiosyncratic error terms follow a generalized extreme value (GEV) distribution. A useful characterization is the implied correlation matrix of errors across products. Ordering products by fuel type g and segment s , the pairwise correlation between the errors of models j and k is

$$\rho_{jk} = \begin{cases} 1 - (\mu_{4,g}/\mu_3)^2, & \text{same } (g, s), \\ 1 - \mu_3^2, & \text{same } g, \text{ different } s, \\ 0, & \text{different } g. \end{cases}$$

Table C.1.1 displays this block structure for a stylized market with two fuel types and two segments, each containing two models.

Table C.1.1: Correlation matrix of idiosyncratic errors in the two-level nested logit

	EV				ICE			
	Seg 1		Seg 2		Seg 1		Seg 2	
	j_1	j_2	j_3	j_4	k_1	k_2	k_3	k_4
j_1	1	$1 - (\mu_4^{EV}/\mu_3)^2$	$1 - \mu_3^2$	$1 - \mu_3^2$	0	0	0	0
j_2	$1 - (\mu_4^{EV}/\mu_3)^2$	1	$1 - \mu_3^2$	$1 - \mu_3^2$	0	0	0	0
j_3	$1 - \mu_3^2$	$1 - \mu_3^2$	1	$1 - (\mu_4^{EV}/\mu_3)^2$	0	0	0	0
j_4	$1 - \mu_3^2$	$1 - \mu_3^2$	$1 - (\mu_4^{EV}/\mu_3)^2$	1	0	0	0	0
k_1	0	0	0	0	1	$1 - (\mu_4^{ICE}/\mu_3)^2$	$1 - \mu_3^2$	$1 - \mu_3^2$
k_2	0	0	0	0	$1 - (\mu_4^{ICE}/\mu_3)^2$	1	$1 - \mu_3^2$	$1 - \mu_3^2$
k_3	0	0	0	0	$1 - \mu_3^2$	$1 - \mu_3^2$	1	$1 - (\mu_4^{ICE}/\mu_3)^2$
k_4	0	0	0	0	$1 - \mu_3^2$	$1 - \mu_3^2$	$1 - (\mu_4^{ICE}/\mu_3)^2$	1

Note: This table displays the implied correlation matrix of the idiosyncratic taste shocks across products in a two-level nested logit. The constraints $0 < \mu_{4,g} \leq \mu_3 \leq 1$ ensure positive definiteness. Within the same (fuel type, segment) cell, the correlation is $1 - (\mu_{4,g}/\mu_3)^2$; within the same fuel type but different segments, it is $1 - \mu_3^2$; across fuel types, it is zero.

C.2 Nesting order: fuel-type first vs. segment first

Our baseline specification nests fuel type above segment: consumers first choose between EV and ICE, then a segment, then a specific model. An alternative is to reverse the ordering: consumers first choose a segment, then a fuel type within that segment.

We compare both orderings by matching two empirical moments from the French second-choice survey (Appendix B.4): the EV retention rate $x_{EV} = 0.814$ (fraction of EV buyers who would choose another EV if their first choice were unavailable) and the within-segment retention rate $x_{seg|EV} = 0.332$.

Table C.2.1 reports the results. The fuel-type-first specification matches both moments exactly ($\hat{\mu}_3 = 0.267$, $\hat{\mu}_{4,EV} = 0.219$). The segment-first specification cannot match both moments simultaneously; at its best fit it underpredicts EV retention and overpredicts segment retention, with a minimum-distance objective an order of magnitude larger than zero.

This result supports the fuel-type-first hierarchy: consumers appear to first decide on the powertrain (EV vs. ICE) and then choose a segment, rather than the reverse.

Hat-algebra derivation of the retention rates. We derive here the closed-form expressions for X_{EV}^m and $X_{s,EV}^m$ used in Table C.2.1. The baseline ordering (fuel $\rightarrow$ segment $\rightarrow$ vehicle) is derived in Section 5.4.1; we focus here on the reverse ordering (segment $\rightarrow$ fuel $\rightarrow$ vehicle).

Under the reverse ordering, segments are the top-level nests, fuel-type nests sit within each segment, and models within each (segment, fuel) cell. Let μ_3 be the scale at the segment $\rightarrow$ fuel level, and $\mu_{4,g}$ the scale at the fuel $\rightarrow$ model level. Inclusive values write

$$IV_{EV|s} = \ln \sum_{k \in (s, EV)} e^{V_k / \mu_{4,EV}}, \quad IV_{ICE|s} = \ln \sum_{k \in (s, ICE)} e^{V_k / \mu_{4,ICE}},$$

$$IV_s = \ln \left(e^{(\mu_{4,EV} / \mu_3) IV_{EV|s}} + e^{(\mu_{4,ICE} / \mu_3) IV_{ICE|s}} \right),$$

and the choice probabilities follow the standard nested-logit form, with $s_s = e^{\mu_3 IV_s} / (1 + \sum_{s'} e^{\mu_3 IV_{s'}})$.

When EV model $j \in (s, EV)$ is removed, only $IV_{EV|s}$ shifts, by $\Delta IV_{EV|s} = \ln(1 - s_{j|s, EV})$. Propagating upward, the affected segment's IV shifts; segments $s' \neq s$ are unchanged in their IVs. Define

$$\Lambda_j = (1 - s_{j|s, EV})^{\mu_{4,EV} / \mu_3}, \quad M_s = 1 - s_{EV|s} + s_{EV|s} \Lambda_j, \quad \Delta_j = M_s^{\mu_3}, \quad \Psi_s = 1 + s_s(\Delta_j - 1).$$

Post-removal: the affected segment share is $s'_s = s_s \Delta_j / \Psi_s$, while unaffected segments rescale by a common factor $s'_{s'} = s_{s'} / \Psi_s$. The post-removal EV share within the affected segment is $s'_{EV|s} = s_{EV|s} \Lambda_j / M_s$. Aggregating across segments,

$$s'_{EV} = \frac{s_{EV} - s_s s_{EV|s}}{\Psi_s} + s'_s s'_{EV|s},$$

and the retention rates take the same functional form as in the baseline:

$$\eta_j^{EV} = \frac{s'_{EV} + s_j - s_{EV}}{s_j}, \quad \eta_j^{(s, EV)} = \frac{s'_s s'_{EV|s} + s_j - s_s s_{EV|s}}{s_j}.$$

The aggregate moments X_{EV}^m and $X_{s,EV}^m$ are then obtained by averaging η_j^{EV} and $\eta_j^{(s, EV)}$ over $j \in EV$ with weights $s_{j|EV}$, exactly as in equation (18).

Two observations. First, $\mu_{4,ICE}$ does not enter the EV-side retention rates: removing an EV model never triggers reallocation among ICE models within a cell, only at the segment-fuel and segment

levels (which use μ_3 and $\mu_{4,EV}$ together with observed shares). Even with fuel-specific $\mu_{4,g}$, the reverse ordering identifies only μ_3 and $\mu_{4,EV}$ from the EV-side moments—the same parameter count as the baseline. The reverse specification’s poor fit reported in Table C.2.1 reflects a misspecification. Second, the reverse ordering uses the unconditional segment shares s_s and the share of EVs in each segment, $s_{EV|s}$, both of which require knowing ICE sales at the segment level—unlike the baseline, which uses only EV-internal shares plus the scalar aggregate s_{EV} .

Table C.2.1: Nesting order comparison: matching second-choice retention rates

Nesting order	Parameters		Implied retention rates		Objective
	μ_3	μ_4	x_{EV}	$x_{seg EV}$	
<i>Empirical targets</i>	—	—	0.814	0.332	—
Fuel $\rightarrow$ Segment $\rightarrow$ Vehicle	0.267	0.219	0.814	0.332	≈ 0
Segment $\rightarrow$ Fuel $\rightarrow$ Vehicle	0.999	0.471	0.589	0.528	0.0887

Note: This table compares two nesting structures by minimum-distance estimation targeting the EV retention rate ($x_{EV} = 0.814$) and the within-segment retention rate ($x_{seg|EV} = 0.332$) from the French second-choice survey (May 2026; Appendix B.4). Parameters are estimated via L-BFGS-B with 50 random starts. The “Objective” column reports the sum of squared deviations from the empirical targets. Closed-form expressions for the retention rates under each nesting order are derived in the hat-algebra block above (and in Section 5.4.1 for the baseline).

C.3 Inference for the nest scale parameters

The estimator of $(\mu_3, \mu_{4,EV})$ described in Section 5.4.1 is exactly identified: two empirical retention moments X_{EV}^o and $X_{s,EV}^o$ pin down two parameters through the hat-algebra map $g(\mu_3, \mu_{4,EV})$. Sampling uncertainty enters only through the survey moments; the market-share inputs are treated as known.

Variance of the empirical moments. For joint inference on $(\hat{\mu}_3, \hat{\mu}_{4,EV})$ we use the strict complete-case sample of EV-first survey respondents whose first- and second-choice segments are both observed ($N = 546$); on this sample the two moments share a common N , so the joint variance-covariance matrix is well defined.²⁵ For each respondent i , define indicators

$$\eta_i^{EV} = \mathbf{1}\{c_i^{(2)} \in EV\}, \quad \eta_i^{s,EV} = \mathbf{1}\{c_i^{(2)} \in EV \text{ and } \text{seg}(c_i^{(2)}) = \text{seg}(c_i^{(1)})\},$$

where $c_i^{(1)}, c_i^{(2)}$ denote the respondent’s first- and second-preferred models. The empirical moments are $\hat{X}^o = \bar{\eta}$, and their variance-covariance matrix is the sample covariance of these indicators di-

²⁵The full EV-first sample has $N = 661$; restricting to first-choice segment observed gives $N = 614$ (the sample used to compute $X_{s,EV}^o = 0.332$ in the body); requiring both segments observed further trims to $N = 546$. Sample means on these three nested samples are: $X_{EV}^o \in \{0.814, 0.813, 0.804\}$ and $X_{s,EV}^o = 0.337$ on the complete-case sample. The point estimates reported in the body use the larger separate samples; the inference below uses the strict complete case throughout.

vided by N :

$$\widehat{V}(\hat{X}^o) = \frac{1}{N} \widehat{\Sigma}_\eta, \quad \widehat{\Sigma}_\eta = \frac{1}{N-1} \sum_{i=1}^N (\eta_i - \bar{\eta})(\eta_i - \bar{\eta})^\top.$$

The implied standard errors are $\widehat{SE}(\hat{X}_{EV}^o) \approx 0.017$ and $\widehat{SE}(\hat{X}_{s,EV}^o) \approx 0.020$, with correlation ≈ 0.32 .

Delta-method standard errors. Let $\hat{\mu} = g^{-1}(\hat{X}^o)$ be the fitted parameters. The Jacobian $G = \partial g / \partial \mu$, evaluated at $\hat{\mu}$, is computed by two-sided finite differences (step $h = 10^{-4}$) on the hat-algebra formulas. Standard delta-method reasoning gives

$$\widehat{V}(\hat{\mu}) = G^{-1} \widehat{V}(\hat{X}^o) (G^{-1})^\top,$$

and under regularity $\hat{\mu} \stackrel{a}{\sim} N(\mu_0, \widehat{V}(\hat{\mu}))$.

Nonparametric bootstrap. We resample the $N = 546$ complete-case respondents with replacement, $B = 1,000$ draws. For each draw, we recompute $(\bar{\eta}^{EV,(b)}, \bar{\eta}^{s,EV,(b)})$, then refit $(\mu_3, \mu_{4,EV})$ under the (RUM) constraint via multistart L-BFGS-B (four random starts plus a warm start at the point estimate). Bootstrap standard errors are the sample standard deviations across draws; 95% percentile intervals are reported. With the 2023 market shares as inputs, no bootstrap draw hits the (RUM) boundary, indicating that survey sampling uncertainty alone is far from challenging (RUM)-feasibility at the baseline.

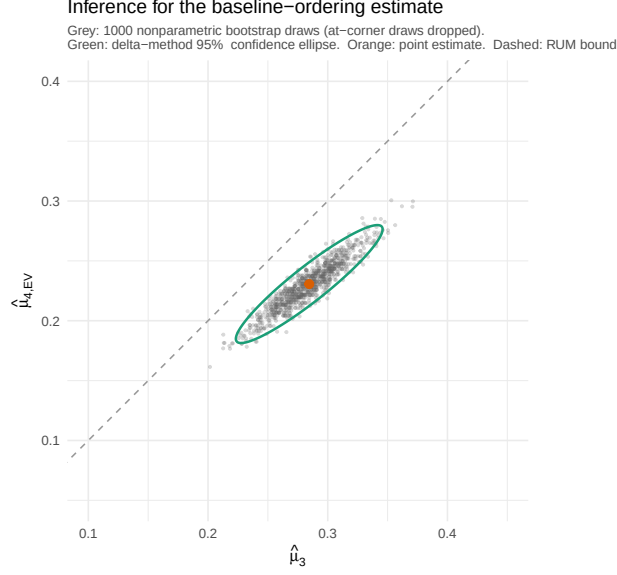
Results. Table C.3.1 reports point estimates, delta-method standard errors, and bootstrap standard errors and percentile intervals. The two methods agree to two decimals, confirming that the linearization of g^{-1} around $\hat{\mu}$ is adequate. The within-parameter correlation is ≈ 0.94 : the data primarily identify the ratio $\hat{\mu}_{4,EV} / \hat{\mu}_3 \approx 0.88$ (through the wedge $X_{EV}^o - X_{s,EV}^o$), with the overall scale identified less tightly. Figure C.3.1 overlays the bootstrap point cloud on the delta-method 95% confidence ellipse; the cloud sits comfortably inside the RUM cone.

Table C.3.1: Inference for $(\hat{\mu}_3, \hat{\mu}_{4,EV})$ under the baseline ordering.

	Point est.	Delta SE	Boot SE	Boot 95% CI lower	Boot 95% CI upper
$\hat{\mu}_3$	0.285	0.025	0.025	0.235	0.335
$\hat{\mu}_{4,EV}$	0.231	0.020	0.020	0.190	0.270

Note: Point estimate from minimum-distance fitting at the strict complete-case sample means ($N = 546$ EV-first respondents with both first- and second-choice segments observed). Delta SE from the sandwich $G^{-1} \widehat{V}(\hat{X}^o) (G^{-1})^\top$ with G computed by two-sided finite differences. Bootstrap based on $B = 1,000$ nonparametric resamples of the same respondents. The 95% CI is the bootstrap percentile interval.

Figure C.3.1: Inference for the baseline-ordering estimate.



Note: Grey: 1,000 nonparametric bootstrap draws of $(\hat{\mu}_3, \hat{\mu}_{4, \text{EV}})$ over the $N = 546$ complete-case respondents. Green: delta-method 95% confidence ellipse. Orange: point estimate at the complete-case sample means. Dashed: (RUM) boundary $\mu_{4, \text{EV}} = \mu_3$.

C.4 Markups in the nested logit model

C.4.1 Analytical formula for markups

Profit-maximising prices in a differentiated-product market. Each firm f owns a set of models $\mathcal{M}_f$ and chooses the price vector $p_f = \{p_j\}_{j \in \mathcal{M}_f}$ to maximise joint profits

$$\Pi_f(p) = \sum_{j \in \mathcal{M}_f} (p_j - c_j) s_j(p), \quad s_j(p) : \text{market share of model } j. \quad (\text{C.19})$$

The first-order condition for every product $i \in \mathcal{M}_f$ is

$$\sum_{j \in \mathcal{M}_f} \frac{\partial s_j}{\partial p_i} (p_j - c_j) + s_i = 0. \quad (\text{C.20})$$

Stacking derivatives in the $J \times J$ matrix $\Delta(p) = \partial s / \partial p'$ and defining the ownership matrix Θ_f with entries $\Theta_{jk} = 1$ if $j, k \in \mathcal{M}_f$ and 0 otherwise, the system can be written compactly as

$$(\Theta_f \odot \Delta(p)) (p - c) = -s(p), \quad (\text{C.21})$$

where $\odot$ denotes the Hadamard product. The optimal mark-up vector is therefore

$$m = p - c = -(\Theta_f \odot \Delta(p))^{-1} s(p). \quad (\text{C.22})$$

For a single-product firm, this collapses to the familiar formula $m_i = -s_i / (\partial s_i / \partial p_i)$.

Closed-form mark-ups under a two-level (energy-segment) nest logit. With a common upper-nest scale μ_3 and group-specific lower-nest scales $\{\mu_{4,g}\}$, the mark-up on product i of firm f can be expressed entirely in terms of observed shares and structural parameters:

$$m_i = \frac{1}{\alpha(1 - S^f)} \frac{1}{\frac{1}{\mu_{4,g}} - \left(\frac{1}{\mu_{4,g}} - \frac{1}{\mu_3}\right)s_{f|s}} \left[1 + \left(\frac{1}{\mu_3} - 1\right) \frac{Z_g^f}{s_g^\circ [1 - (\frac{1}{\mu_3} - 1)Z_g^f/s_g^\circ]} \right], \quad (\text{C.23})$$

where: s_g° is the share of engine-type g in total sales; $s_{f,s}$ the share of firm f in segment s ; $s_{f|s} = s_{f,s}/s_s^\circ$ the firm-segment share; $D_s^f = 1/\mu_{4,g} + (\frac{1}{\mu_3} - \frac{1}{\mu_{4,g}})s_{f|s}$; $Z_g^f = \sum_s s_{f,s}/D_s^f$; $S^f = \sum_g \frac{Z_g^f}{1 + (1 - \frac{1}{\mu_3})Z_g^f/s_g^\circ}$.

Equation (C.23) shows that mark-ups are a function of market shares and the parameters $\theta = \{\alpha, \mu_3, \mu_{4,EV}, \mu_{4,ICE}\}$. In particular, all products of a given firm share the same mark-up within a segment-engine nest (s, g) , a property exploited in the estimation procedure. This property has been shown by Hottman et al. (2016) to hold for nested CES demand system.

C.4.2 Iterative procedure

Here, we describe the iterative process that can be used to recover $\alpha/\mu_{4,g}$ by controlling for markups as given by equation C.23, in the spirit of Head and Mayer (2021) and Breinlich et al. (2024).

We start with a guess $b^{(n)} = \{\alpha^{(n)}, \mu_{4,ICE}^{(n)}\}$ and estimates for $\{\mu_3, \mu_{4,EV}\}$.

1. Use equation C.23 to compute markups $m_{jt}^{(n)}$.
2. Construct a new dependent variable $y_{jt}^{(n)} = \ln s_{j|g,t} - \alpha^{(n)}/\mu_{4,g}^{(n)} m_{jt}^{(n)}$, where $g = \text{ICE}, \text{EV}$.
3. Run regression using $y_{jt}^{(n)}$ as dependent variables and recover estimates from which we can get $b^{(n+1)}$.
4. Start from step (1) until $\text{dist}(b^{(n)}, b^{(n+1)}) < \text{tol}$.

C.5 Welfare tables under Bertrand pricing

Tables C.5.1 and C.5.2 present the counterparts of the main welfare tables (Tables 11 and 12) under multiproduct Bertrand pricing. Allowing markups to adjust endogenously has a modest quantitative impact: the AVPF of the actual reform moves from 1.07 to 1.09 (without fiscal externalities) and from 0.75 to 0.76 (with fiscal externalities). For the budget-neutral reform, the main difference is a larger consumer surplus loss (−85M vs. −61M) as winning firms extract higher markups from their improved competitive position.

Table C.5.1: AVPF of the actual reform (Bertrand pricing).

	W/o fiscal ext.	W/ fiscal ext.
Numerator		
Global: $\Delta CV + \Delta EX_{TTW} + \Delta EX_{LCA} + \Delta PS$	-258	-258
Local: $\Delta CV + (1 - t) \theta_{FR} \Delta PS_W$	-218	-218
Denominator ($-\Delta G$)		
Bonus savings	-236	-236
Fuel tax (TICPE)	—	-94
Corporate tax on FR PS	—	-10
Global AVPF	1.09	0.76
Local AVPF	0.92	0.64

Note: Multiproduct Bertrand pricing. Same structure as Table 11. Fuel tax under Bertrand is scaled from the fixed-price estimate using the Bertrand/fixed-price TTW ratio. See Table 11 notes for parameter definitions.

Table C.5.2: Budget-neutral reform: welfare decomposition (Bertrand pricing).

Component	M EUR
Consumer surplus (ΔCV)	-84
Environmental TTW (SCC=200)	-25
Environmental LCA (SCC=200)	+52
French producer surplus	+105
European producer surplus	+46
Asian producer surplus	-74
Rest of World producer surplus	-26
Total ΔPS	+51
Net welfare (global)	-6
Local welfare ($\Delta CV + (1 - t) \theta_{FR} \Delta PS_W + t \Delta PS_{FR}$)	-47
Local welfare + env.	-20

Note: Multiproduct Bertrand equilibrium. Same structure as Table 12. $\theta_{FR} = 20\%$, $t = 25\%$. See Table 12 notes for parameter definitions.

C.6 Parameters used for the cost benefit analysis

We detail here the main input to the cost-benefit analysis.

C.6.1 Parameters

Social cost of carbon is set at €200 per ton. Tank to wheel emissions are retrieved from the WLTP number. Following findings from the EU commission (European Commission, 2024) they are inflated by a factor of 1.2 for gasoline and diesel engines including non-plugin hybrids and by a factor of 3.5 for hybrids. We assume vehicles drive 180 000 km during their lifetime.

C.6.2 Method used to compute manufacturing-phase emissions

For every vehicle we estimate the embodied CO₂ of the **manufacturing stage only**. We closely follow the methodology employed by ADEME and enrich it with parameters from the International Energy agency (ref).

$$CO_{2,mfg} = CO_{2,materials} + CO_{2,assembly} + CO_{2,battery} + CO_{2,transport}$$

Inputs.

- Curb mass m (kg) and engine type
- Final-assembly country (share when multiple plants)
- Battery sourcing region (share, when relevant)

1. Materials

$$CO_{2,materials} = CO_{2,ferrous} + CO_{2,aluminum} + CO_{2,other}$$

Using the ADEME methodology for the calculation of the environmental score, we utilize the following country and material specific emission factors λ . The weight w of the material is divided by a loss factor δ that takes account of the fact that material is lost during production. The loss factor is $\delta = 0.3$ for ferrous metals and aluminum and $\delta = 0$ for other materials.

$$CO_{2,material}^{origin} = \lambda_{material}^{origin} * \frac{w_{material}}{1 - \delta_{material}}$$

1. Ferrous metals

	$\lambda_{ferrous}$
France	1.4
Europe (avg)	1.4
Mainland China	2.0
Asia w/o China	1.9
Rest of World	2.0

2. Aluminum

	$\lambda_{aluminum}$
France	8.6
Europe (avg)	8.6
Mainland China	20
China w/o Asia	18.5
Rest of World	18.5

3. Other materials

	λ_{other}
France	4.6
Europe (avg)	4.6
Mainland China	5
China w/o Asia	5
Rest of World	5

Our data does not allow us to identify the exact mix of materials used in a car. Besides the characteristics like engine-type, power, etc. we do however have the weight of the vehicle (including the battery). We assume that the battery of an EV makes up about 20% of its weight and take this into account when calculating the materials' contribution to CO₂ emissions. To map the weight of a car into the weight of its different materials, we rely on the methodology underlying the International Energy Agency's (IEA) Electric Vehicle Life Cycle Assessment Calculator (who use the GREET database managed by the US Department of Energy's Argonne National Laboratory). They provide the following average material decomposition by weight (excluding the EV battery) for electric vehicles, hybrids and combustion engine vehicles:

Material	EV	PHEV	ICE
Steel	66%	61%	63%
Aluminum	13%	14%	13%
Copper	4%	6%	2%
Glass	2%	2%	2%
Plastic	12%	13%	16%
Rubber	3%	3%	4%

Using this, we approximate the weight of each material for the above CO₂ calculation as:

$$w_{ferrous} = w_{vehicle} \times s_{steel}^{power-train}$$

$$w_{aluminum} = w_{vehicle} \times s_{aluminum}^{power-train}$$

$$w_{other} = w_{vehicle} \times (s_{copper}^{power-train} + s_{glass}^{power-train} + s_{plastic}^{power-train} + s_{rubber}^{power-train})$$

2. Assembly & Intermediary Energy Usage

The second component, CO_{2,assembly} covers the emissions linked to the energy used in intermediate transformations of materials and at the assembly of the vehicle. The country specific factor λ below, is again multiplied with a vehicles weight (excluding the weight of the battery).

$$CO_{2,assembly} = w_{vehicle} \times \lambda_{assembly}^{origin}$$

	$\lambda_{assembly}$
France	0.58
Europe (avg.)	0.76
Mainland China	1.60
Asia w/o China	1.56
Rest of World	1.03

3. Battery

The third component, CO_{2,battery} determines the emissions linked to the battery production by relating a battery's capacity in KWh to country specific emissions factors. This component hence applies to electric vehicles only. Using information on vehicles from the ADEME data that were perfectly mapped into the battery data, we use a simple regression of capacity on weight and power

in kilowatt to predict a vehicles capacity (adjusted R-squared of 0.90).

$$CO_{2,battery} = kwh_{vehicle} \times \lambda_{battery}^{origin}$$

	$\lambda_{battery}$
France	53
Europe (avg.)	53
Mainland China	68
Asia w/o China	63
Rest of World	55
Unknown	60

4. Transport to France

The fourth component, $CO_{2,transport}$, relates the distance of the place of assembly from France to emissions factors of different modes of transport. The weight (including battery) is here measured in tons.

$$CO_{2,transport} = (w_{vehicle}/1000) \times \lambda_{mode}^{origin} \times Distance_{km}$$

	Distance (km)	Mode	λ_{mode}^{origin}
France	300	Road	0.208
Europe	1000	Road	0.256
Mainland China	20000	Sea	0.101
Asia w/o China	20000	Sea	0.101
Rest of World	9000	Sea	0.101